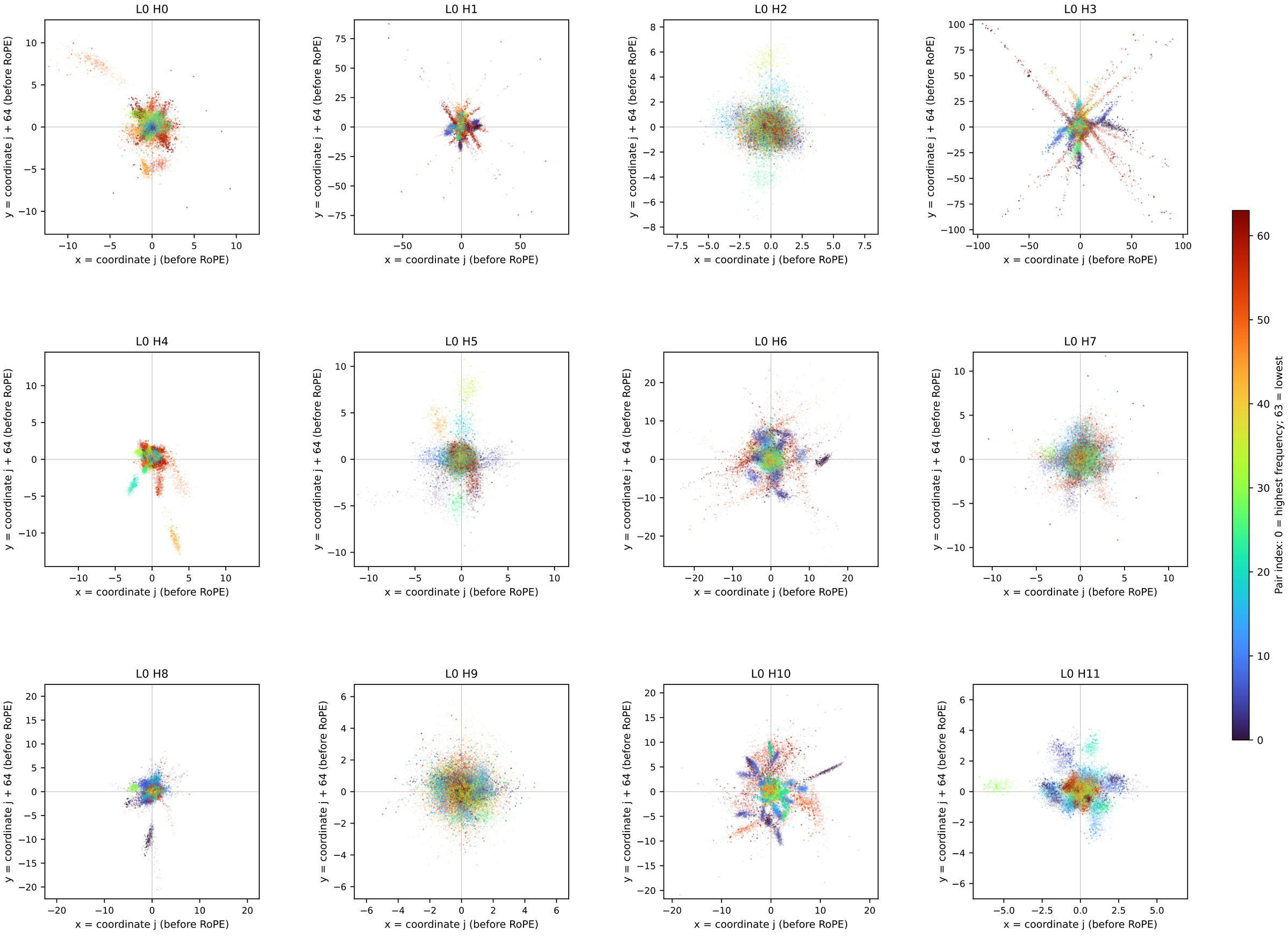
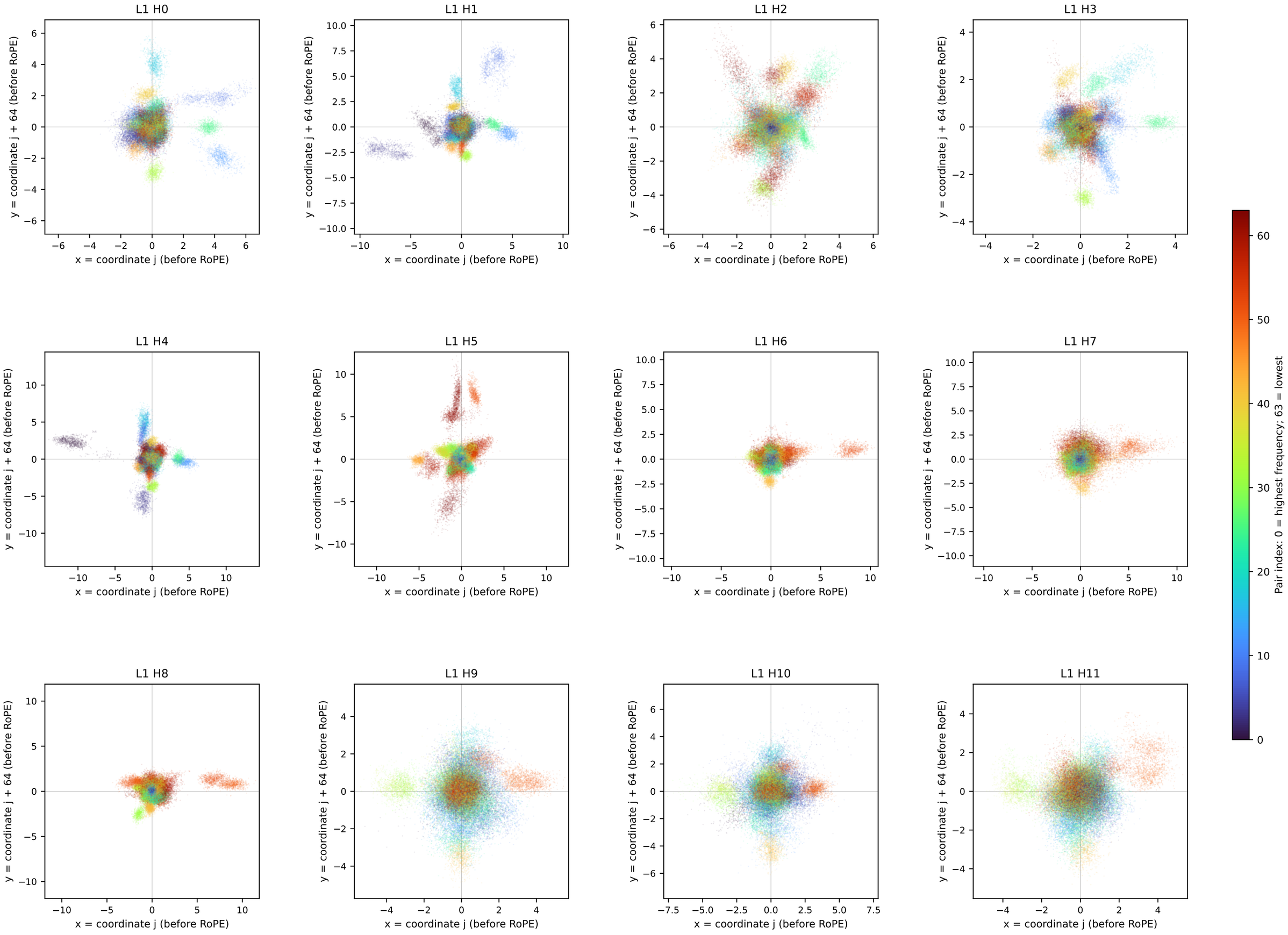
Qwen queries · layer 0 · axes fitted separately for each head



Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

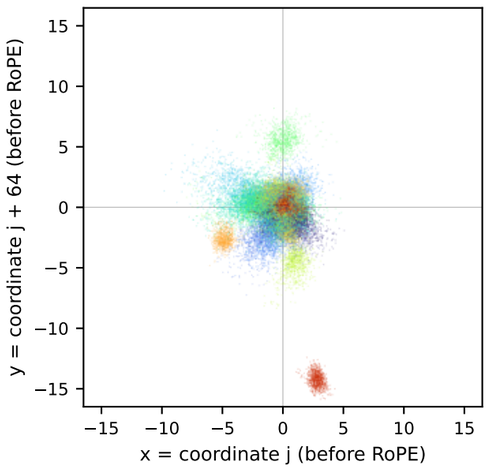
Qwen queries · layer 1 · axes fitted separately for each head



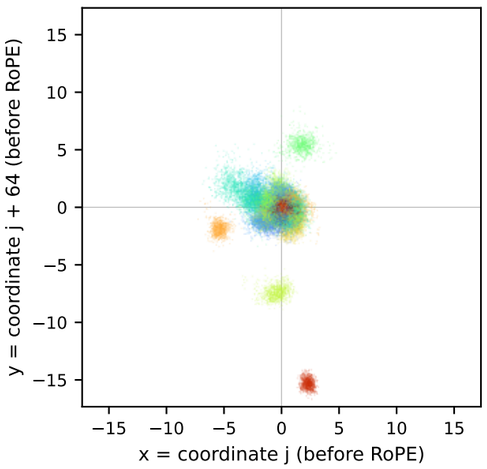
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 2 · axes fitted separately for each head

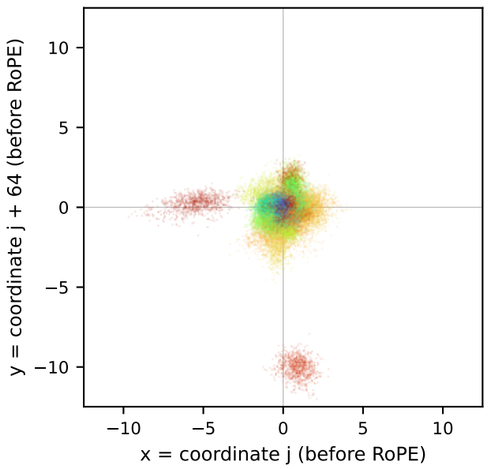
L2 H0



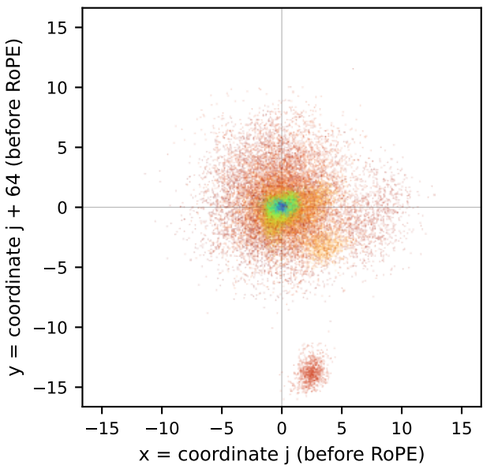
L2 H1



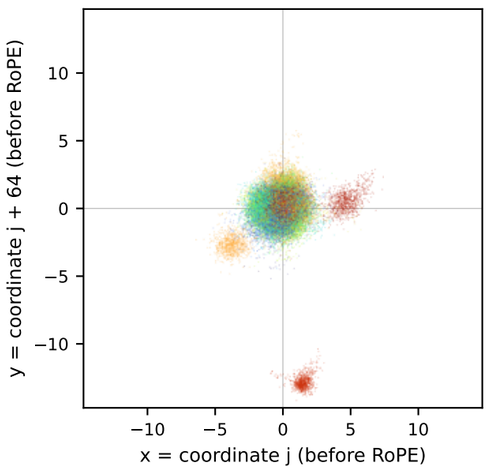
L2 H2



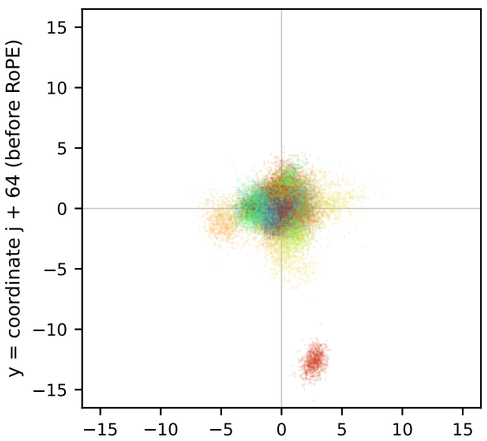
L2 H3



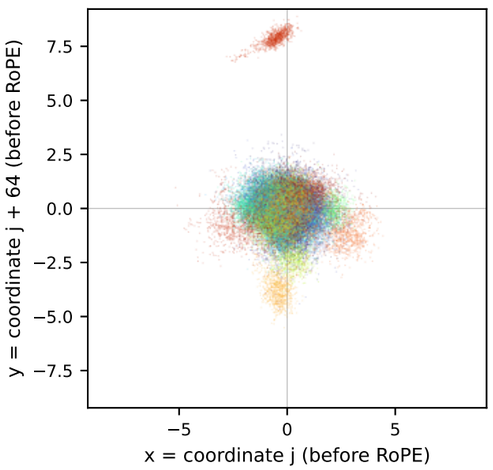
L2 H4



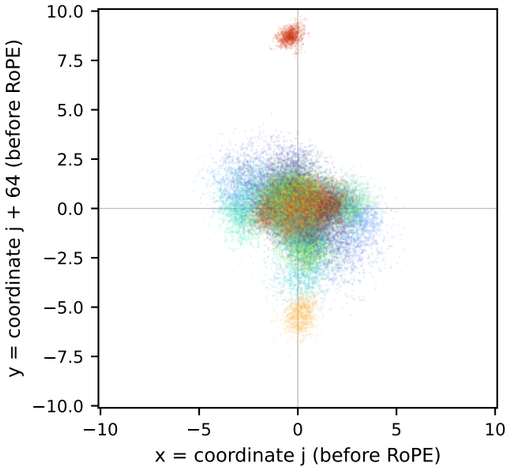
L2 H5



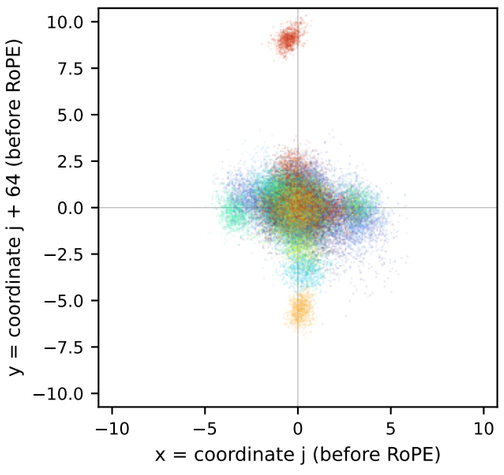
L2 H6



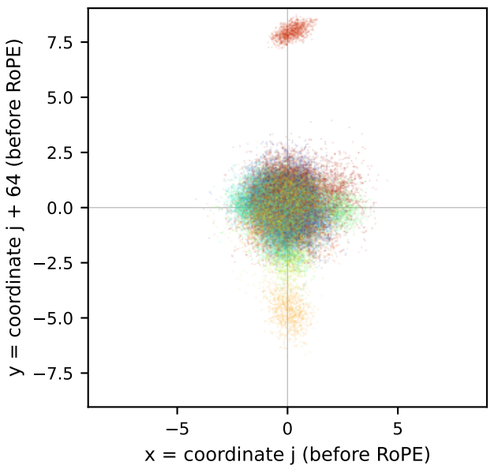
L2 H7



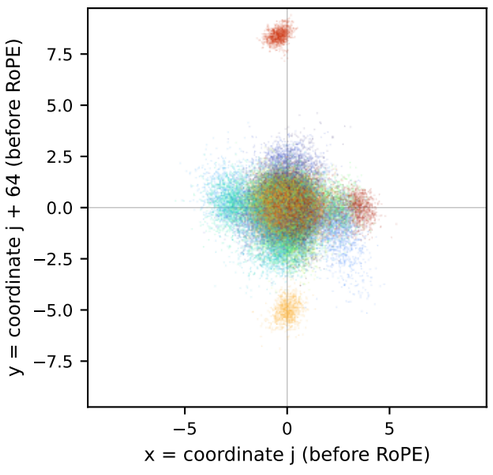
L2 H8



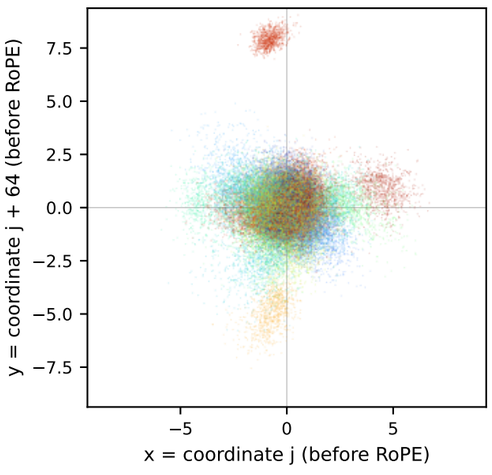
L2 H9

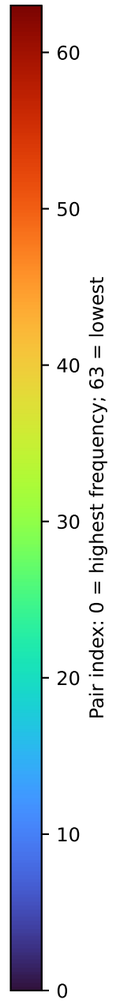


L2 H10



L2 H11

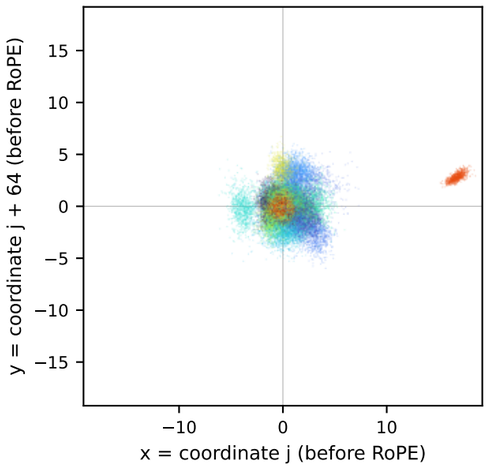




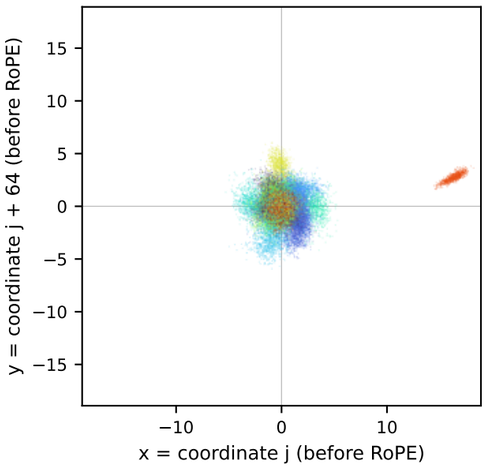
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 3 · axes fitted separately for each head

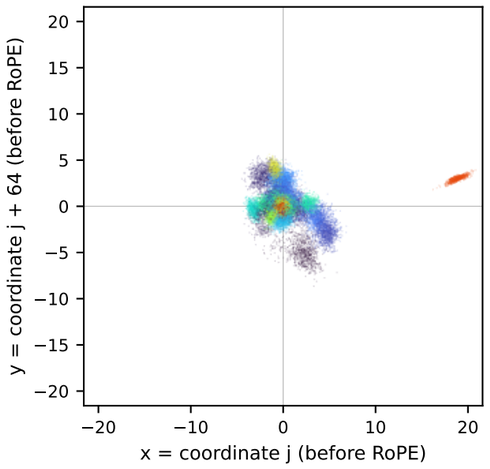
L3 H0



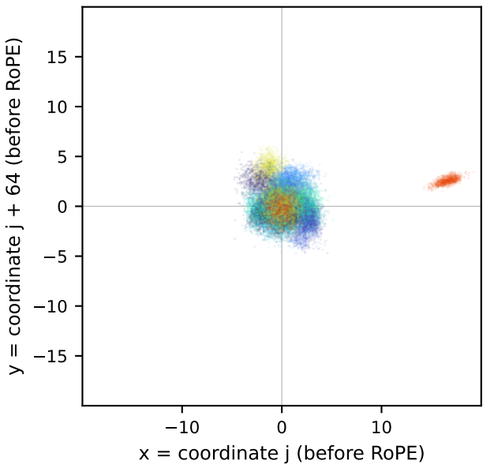
L3 H1



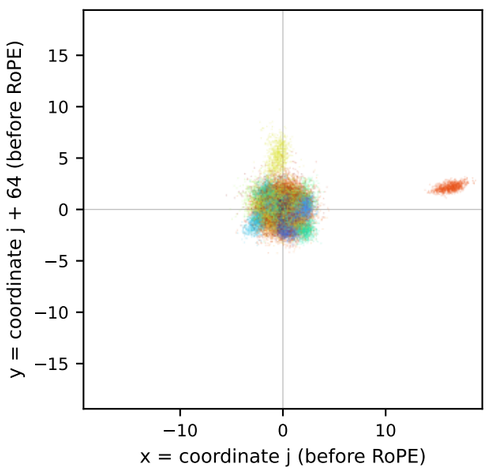
L3 H2



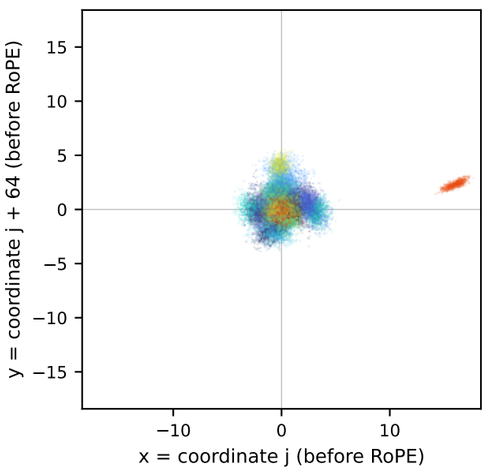
L3 H3



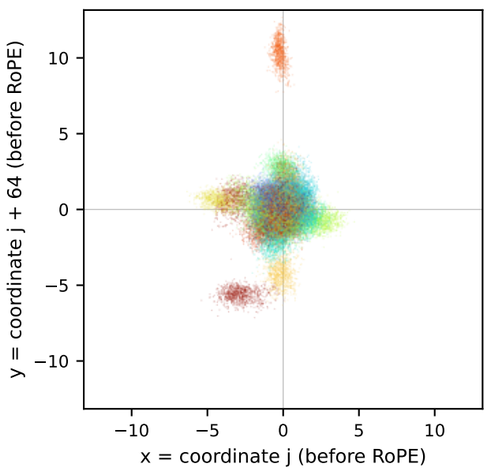
L3 H4



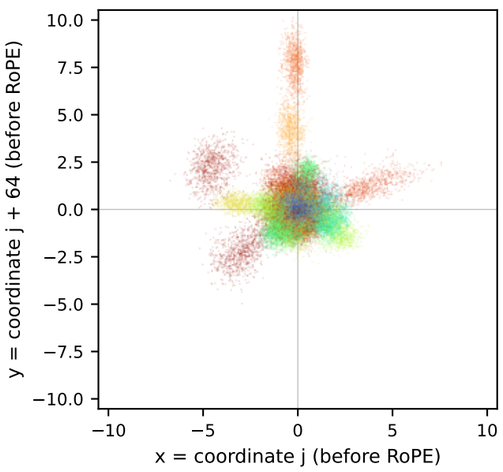
L3 H5



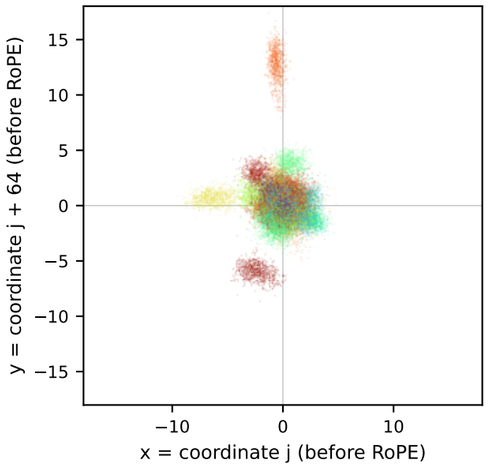
L3 H6



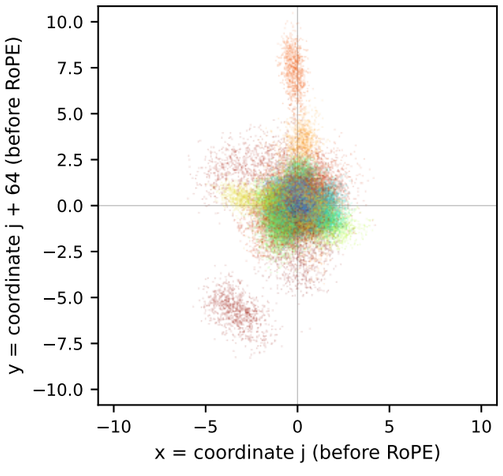
L3 H7



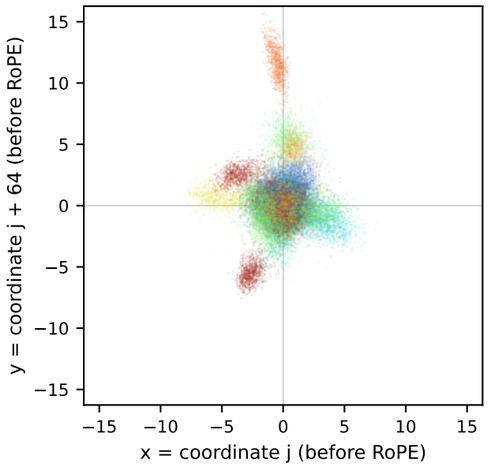
L3 H8



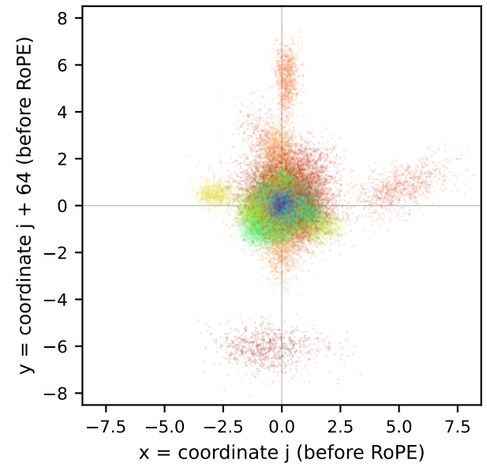
L3 H9

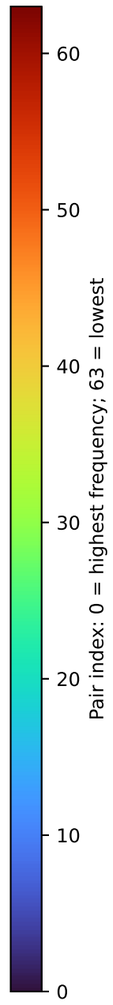


L3 H10



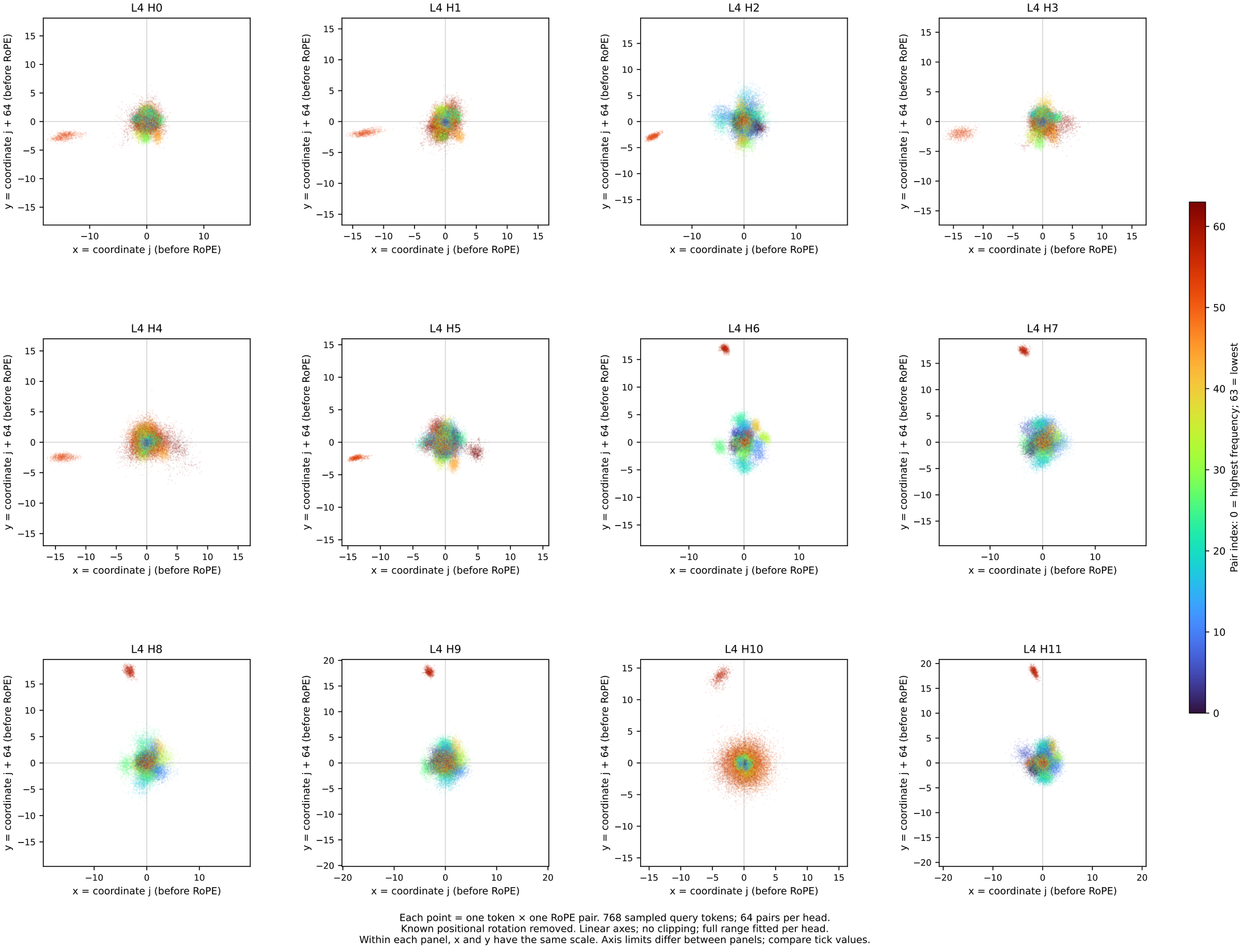
L3 H11





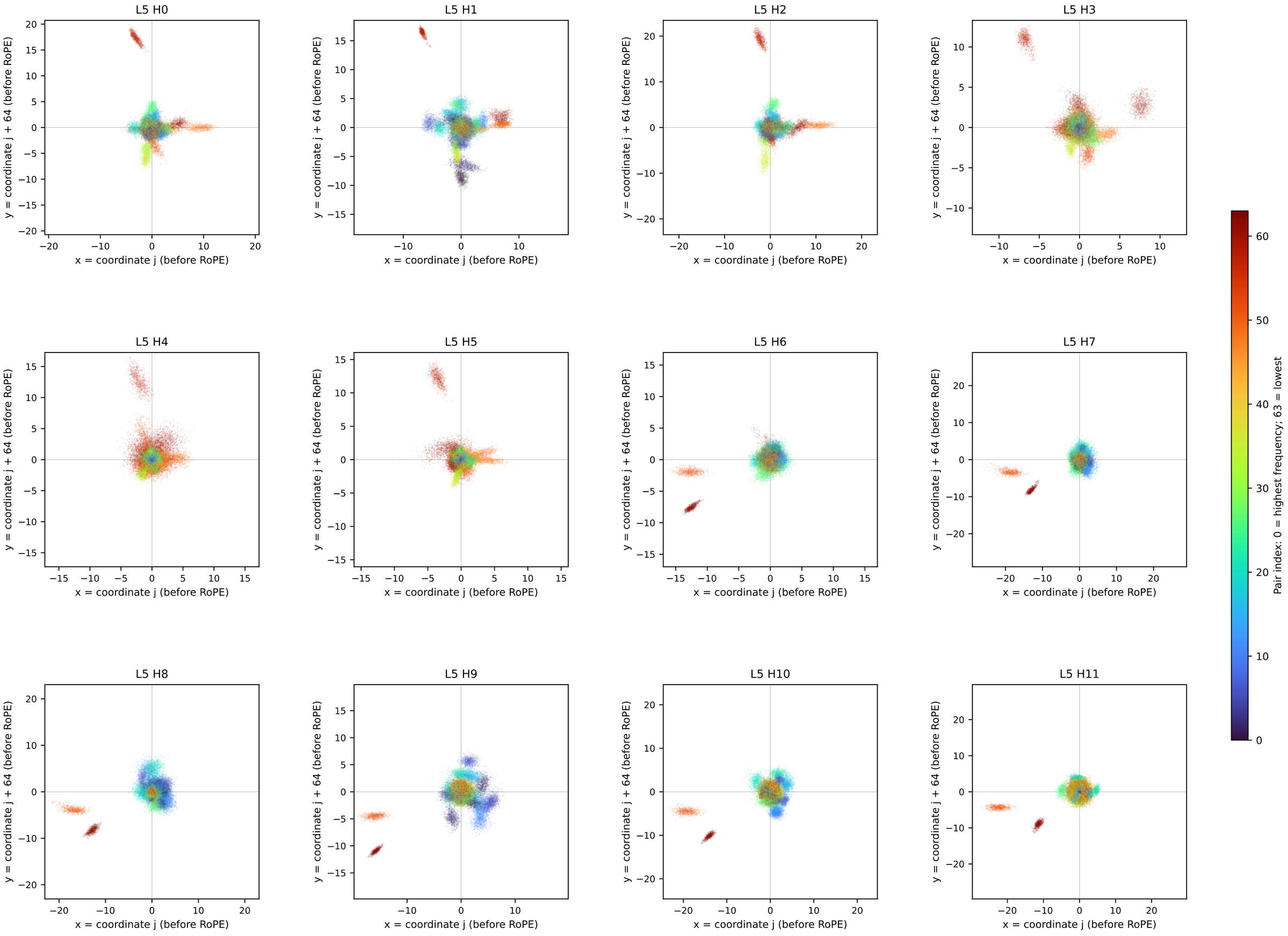
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 4 · axes fitted separately for each head



Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

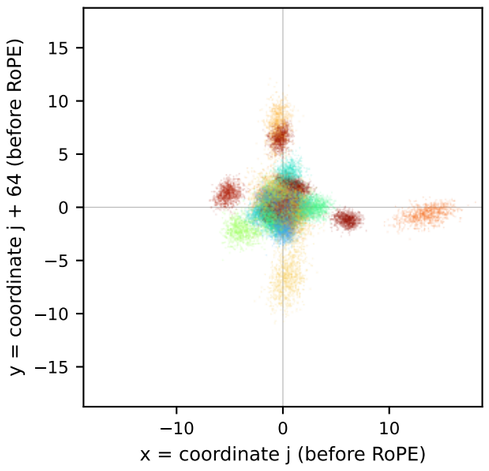
Qwen queries · layer 5 · axes fitted separately for each head



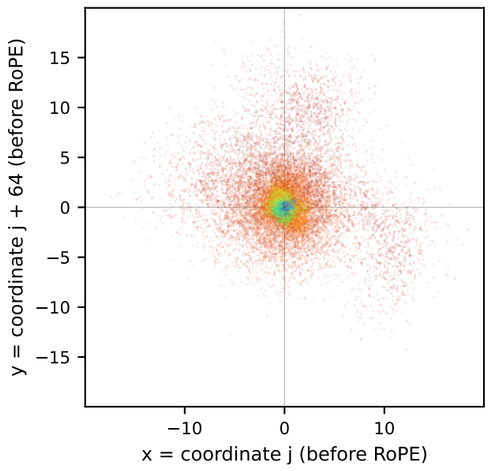
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 6 · axes fitted separately for each head

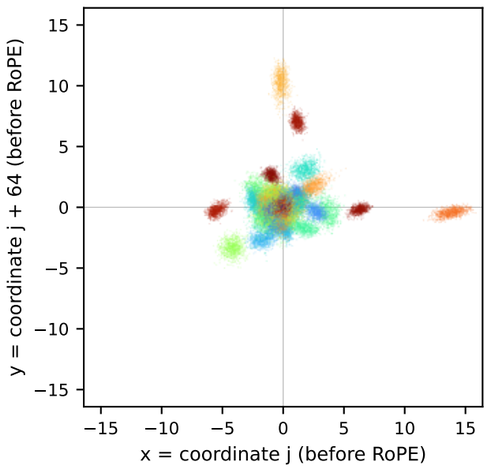
L6 H0



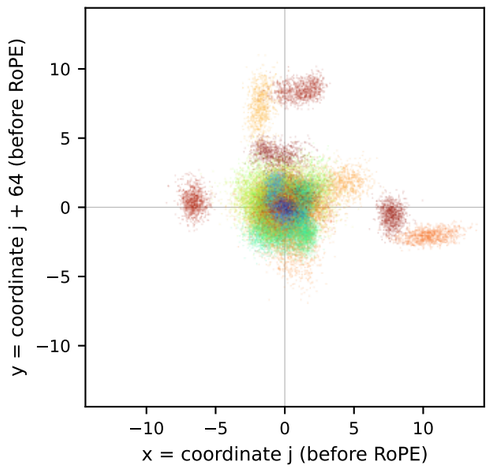
L6 H1



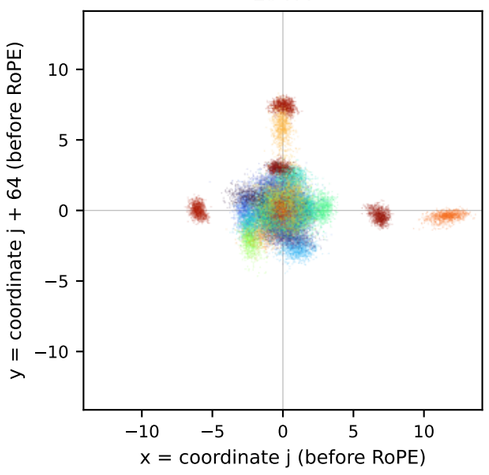
L6 H2



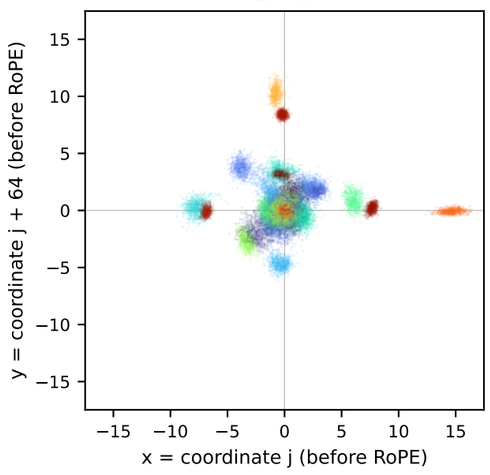
L6 H3



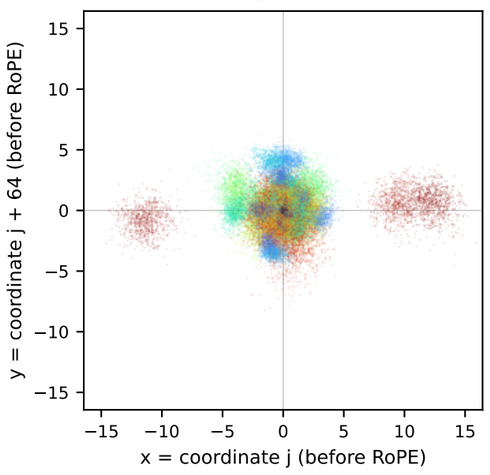
L6 H4



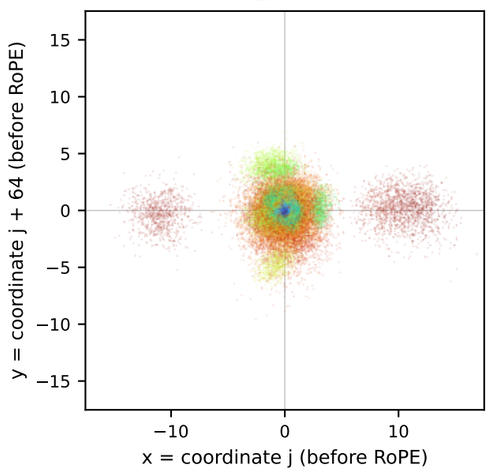
L6 H5



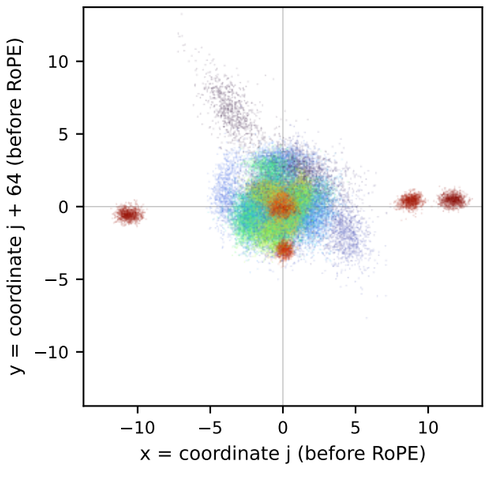
L6 H6



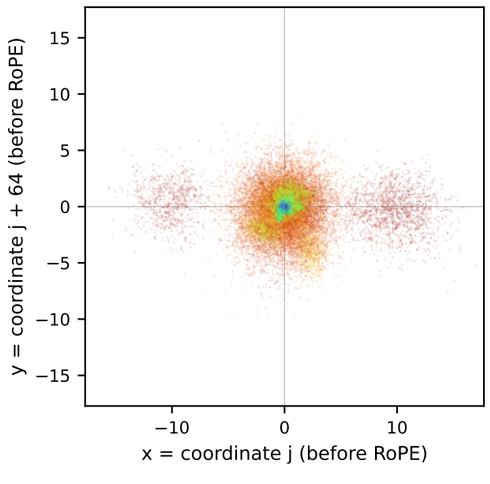
L6 H7



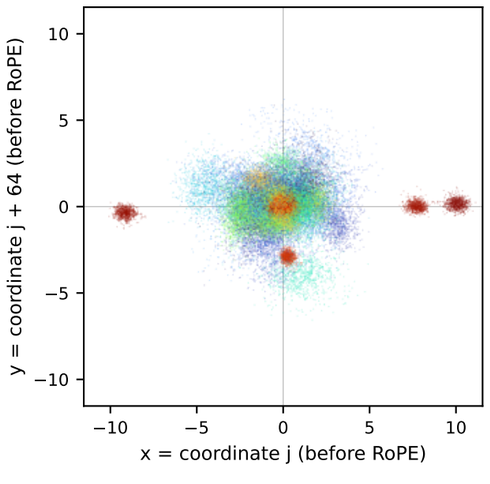
L6 H8



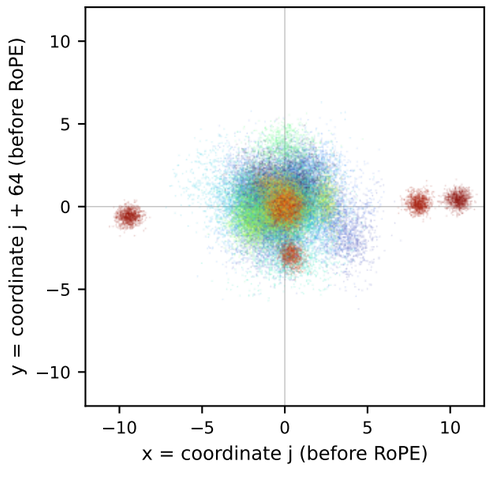
L6 H9

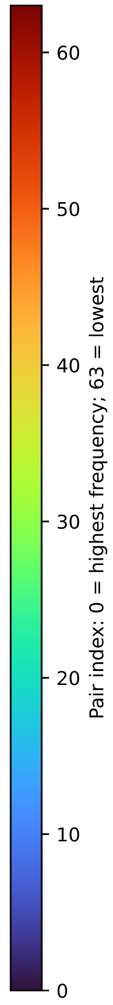


L6 H10



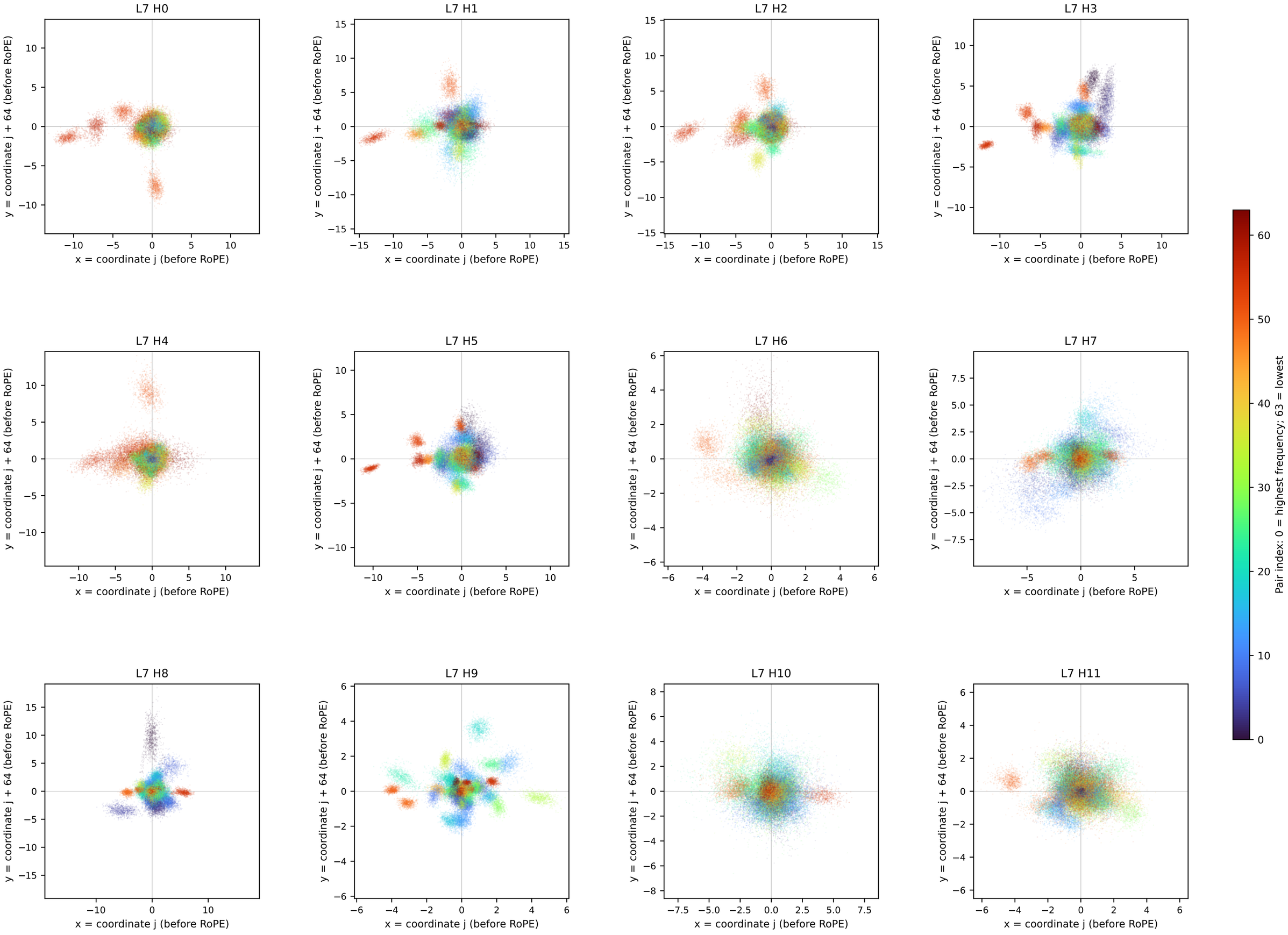
L6 H11





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Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

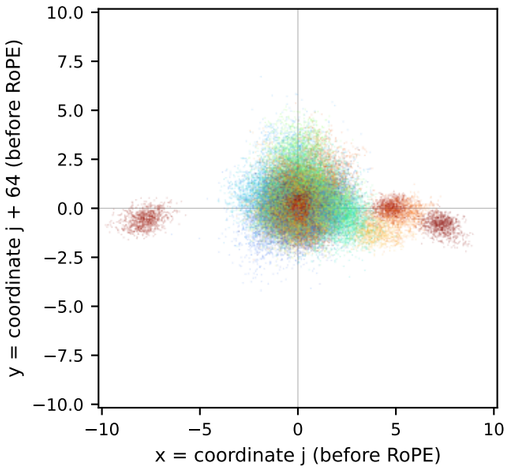
Qwen queries · layer 7 · axes fitted separately for each head



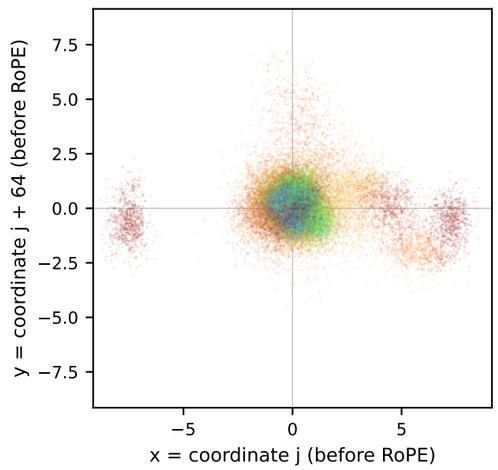
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 8 · axes fitted separately for each head

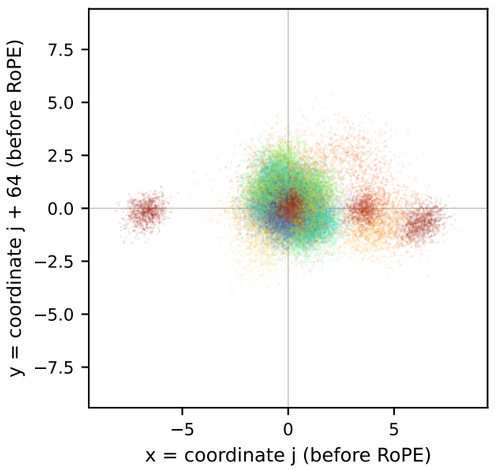
L8 H0



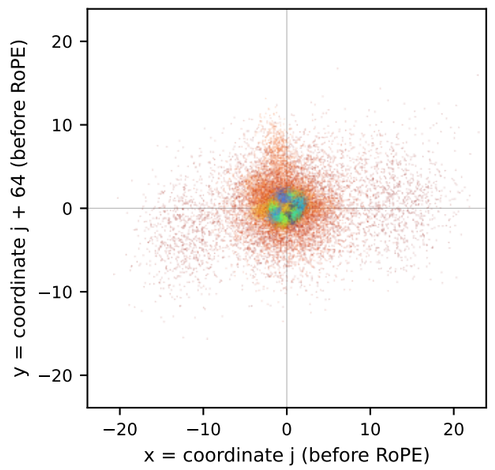
L8 H1



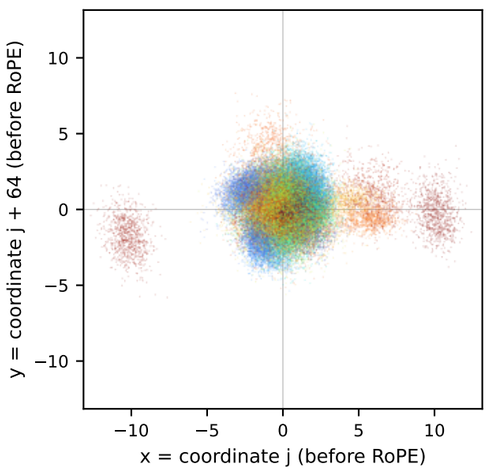
L8 H2



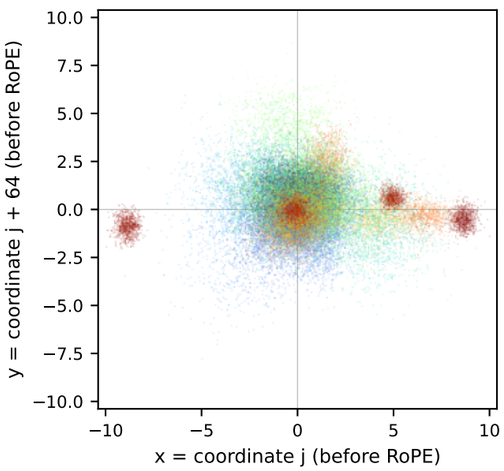
L8 H3



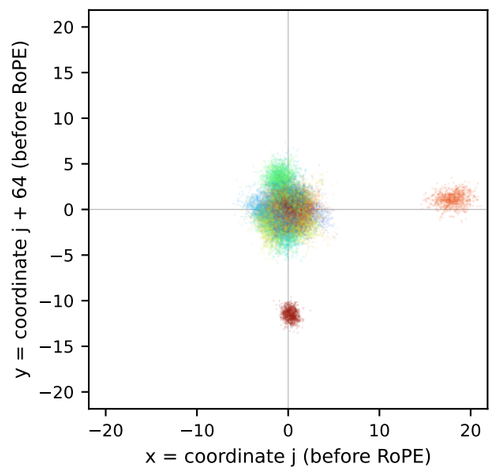
L8 H4



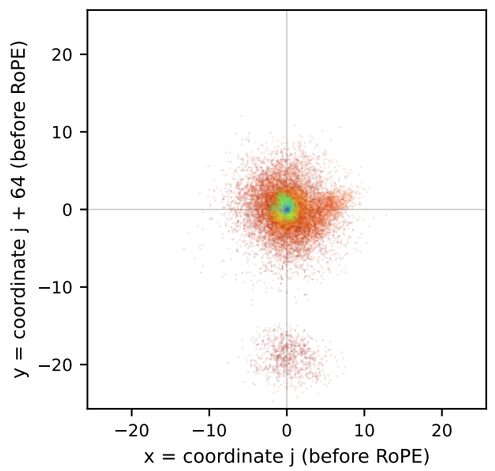
L8 H5



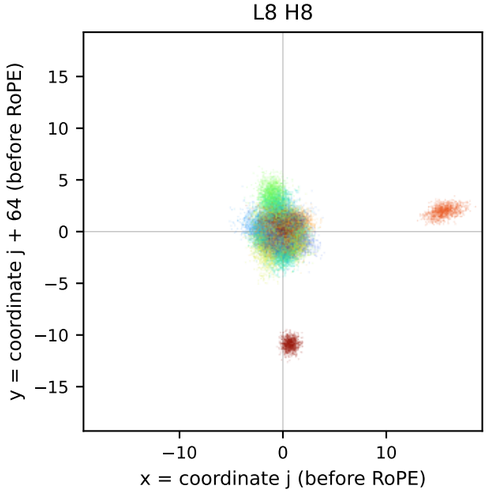
L8 H6



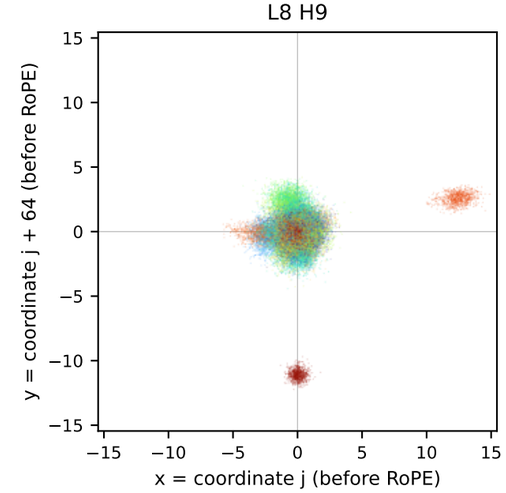
L8 H7



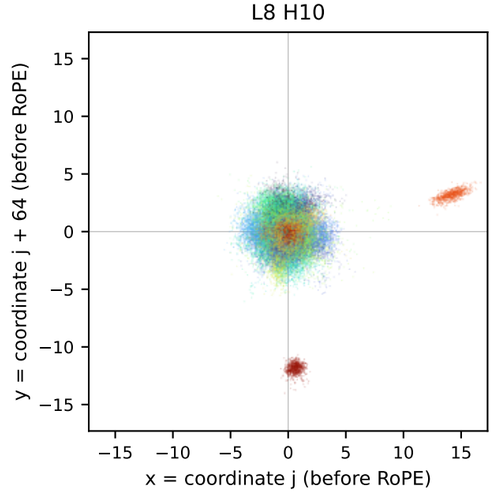
L8 H8



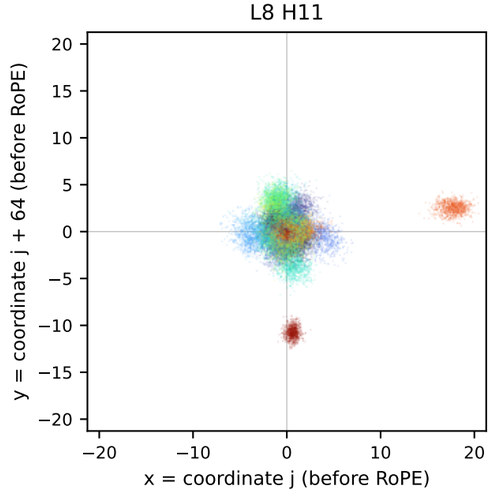
L8 H9

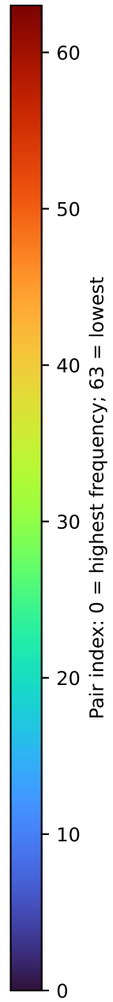


L8 H10



L8 H11

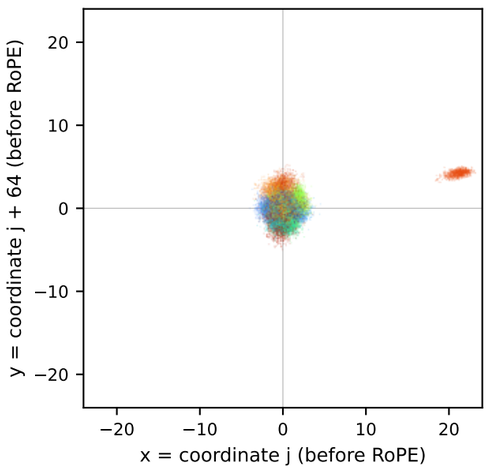




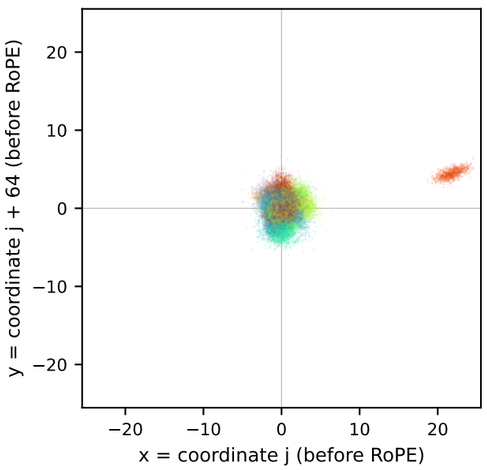
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 9 · axes fitted separately for each head

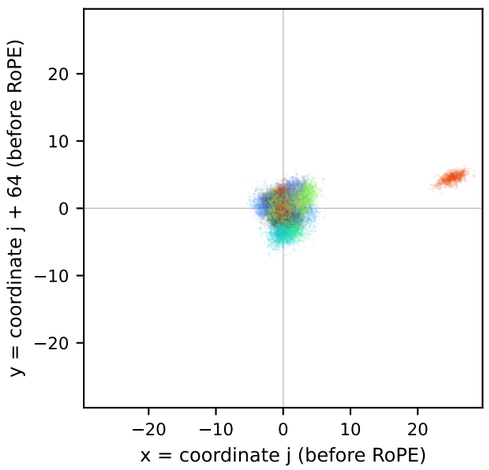
L9 H0



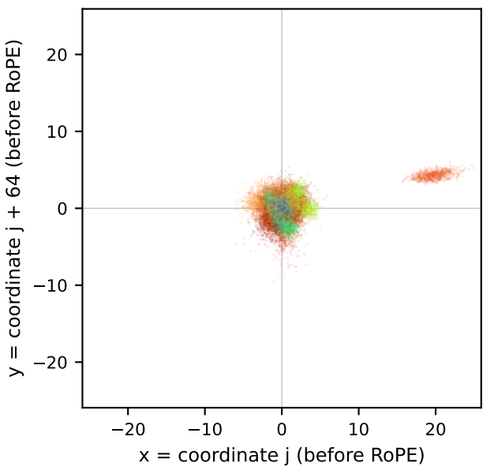
L9 H1



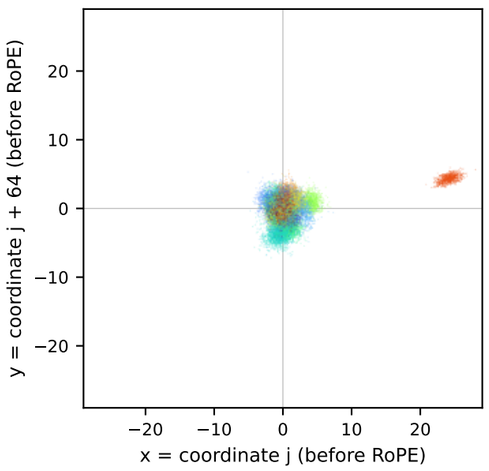
L9 H2



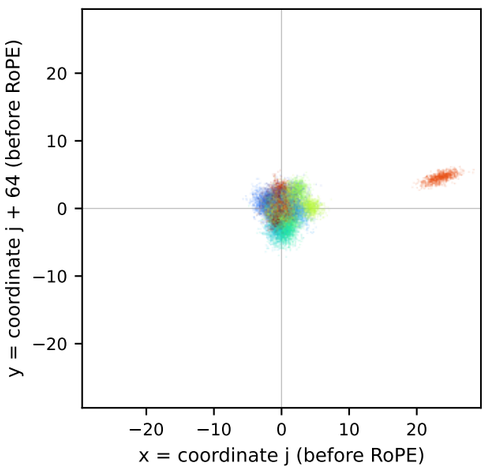
L9 H3



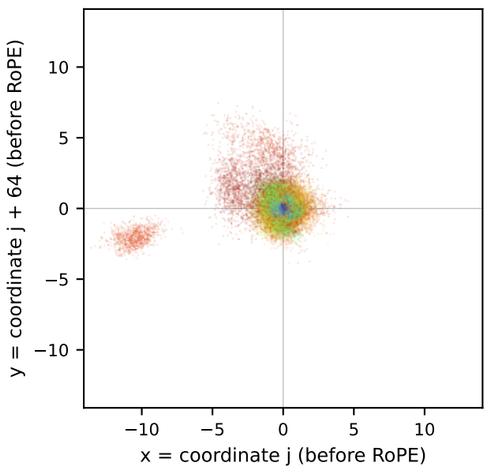
L9 H4



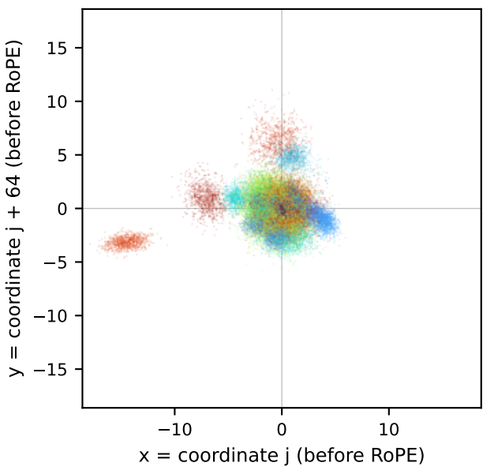
L9 H5



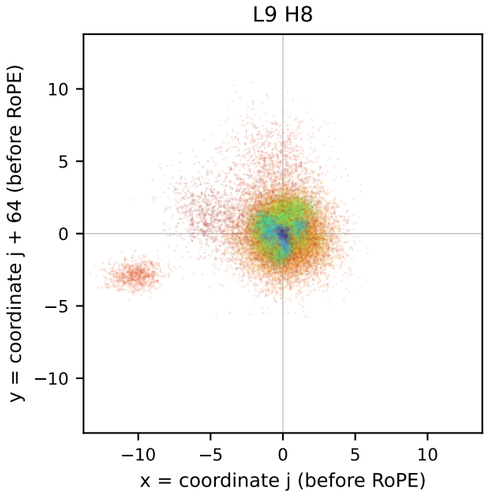
L9 H6



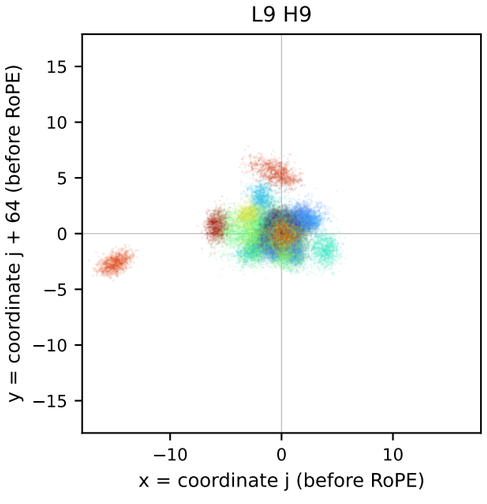
L9 H7



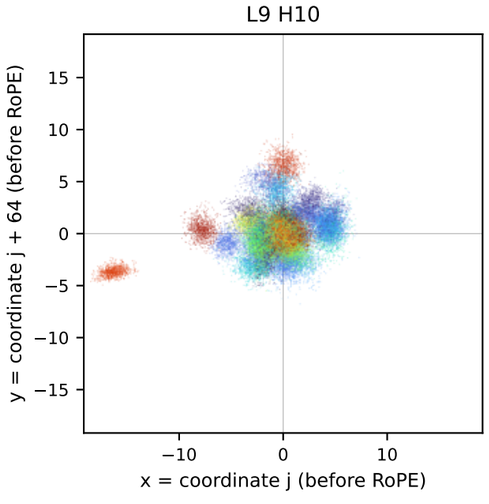
L9 H8



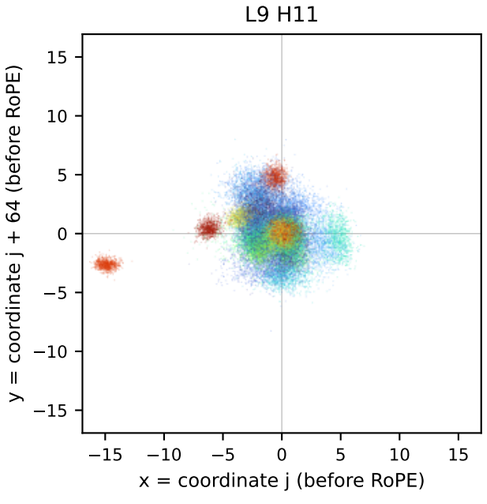
L9 H9

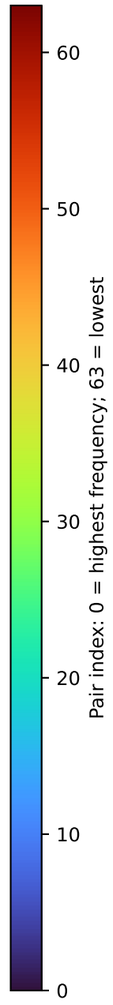


L9 H10



L9 H11

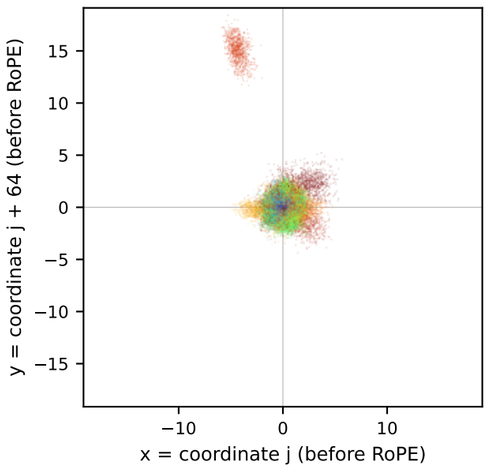




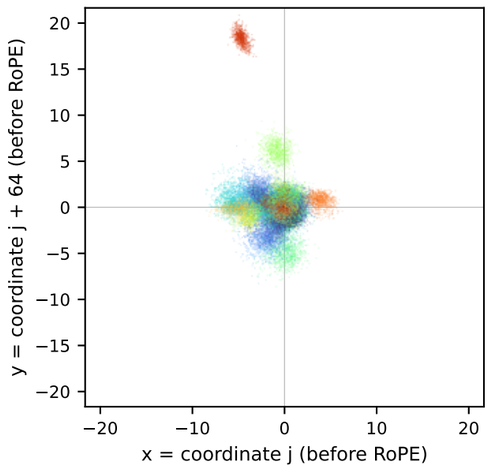
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 10 · axes fitted separately for each head

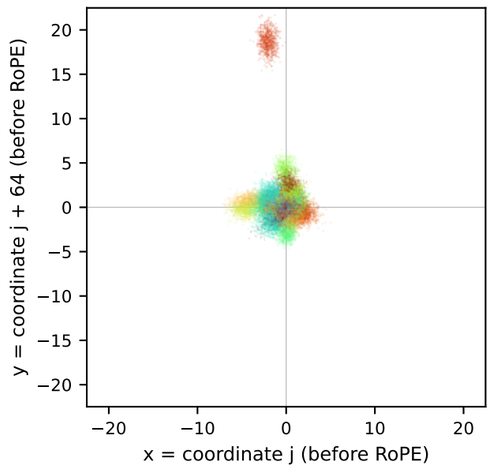
L10 H0



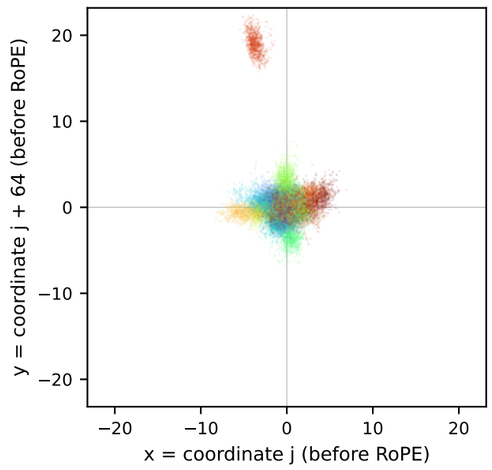
L10 H1



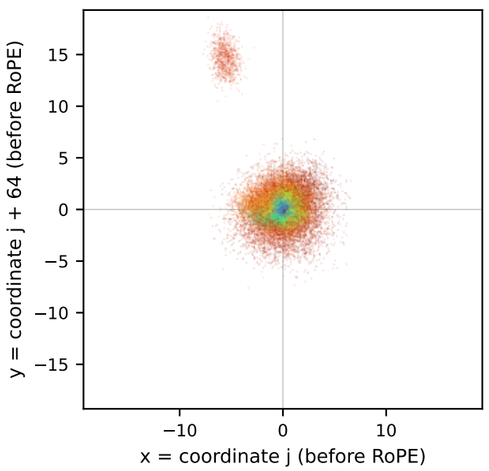
L10 H2



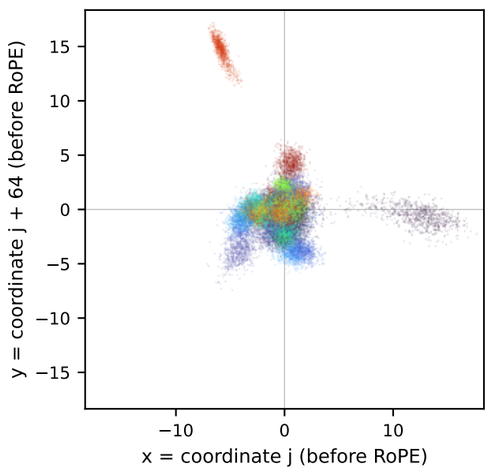
L10 H3



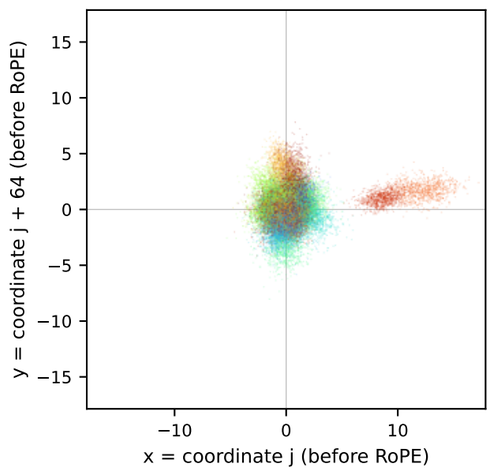
L10 H4



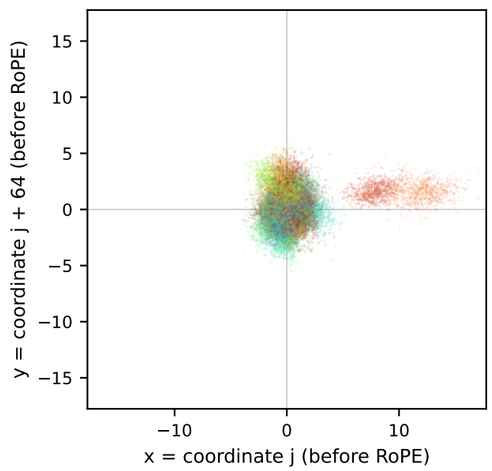
L10 H5



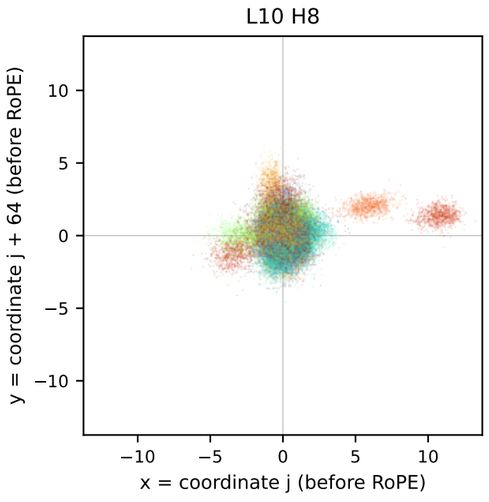
L10 H6



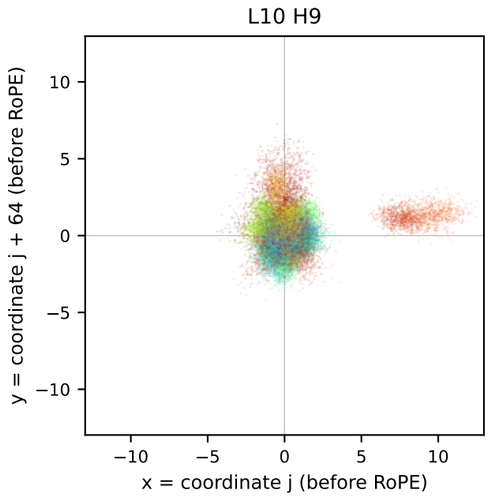
L10 H7



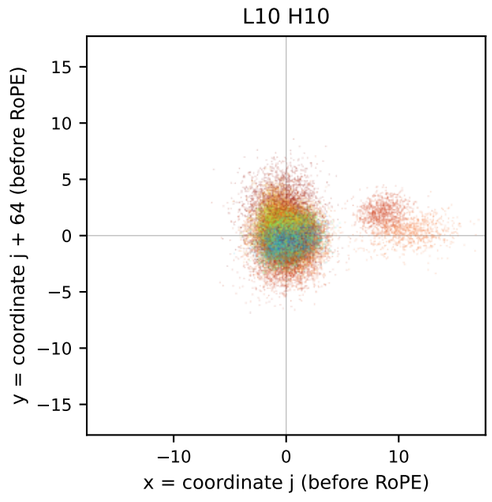
L10 H8



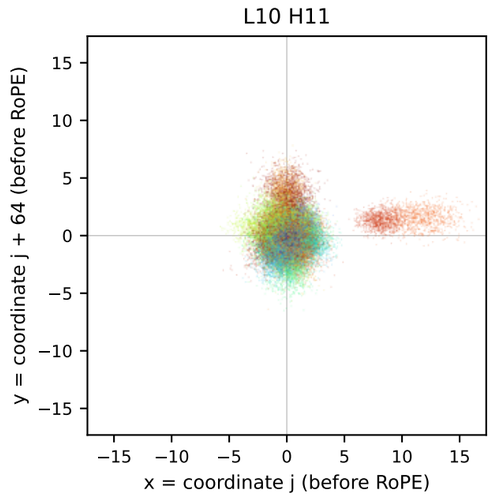
L10 H9

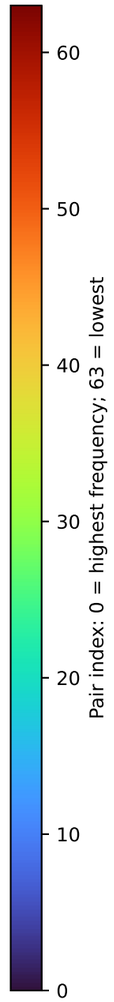


L10 H10



L10 H11

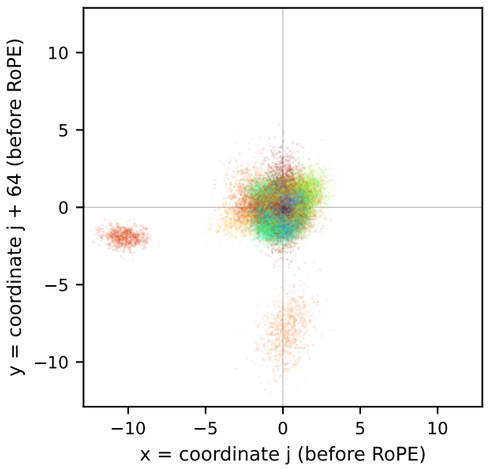




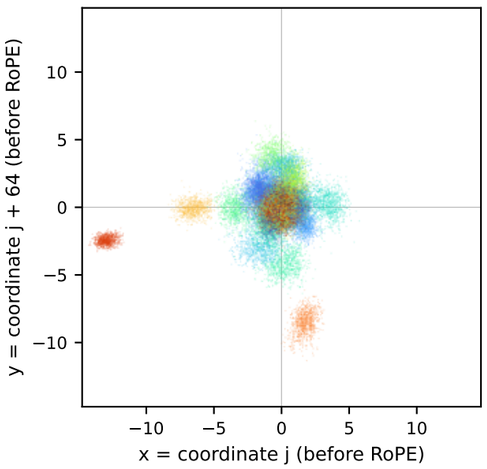
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 11 · axes fitted separately for each head

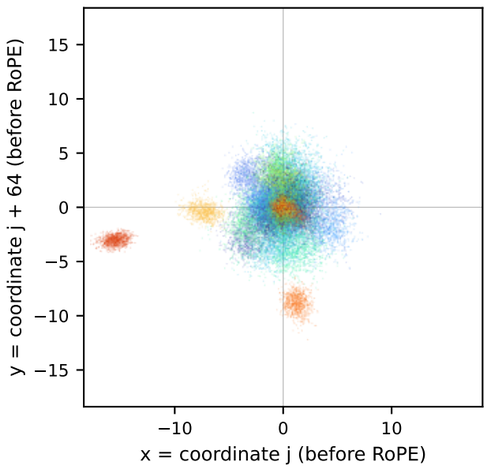
L11 H0



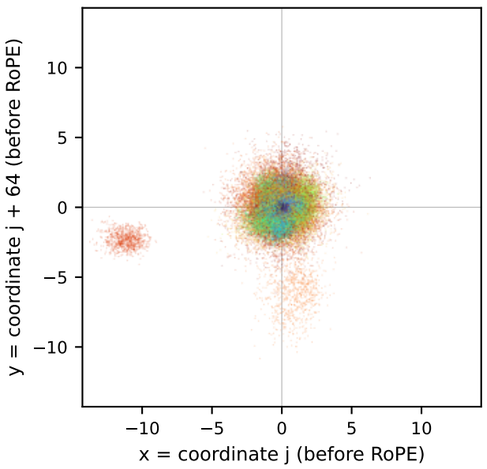
L11 H1



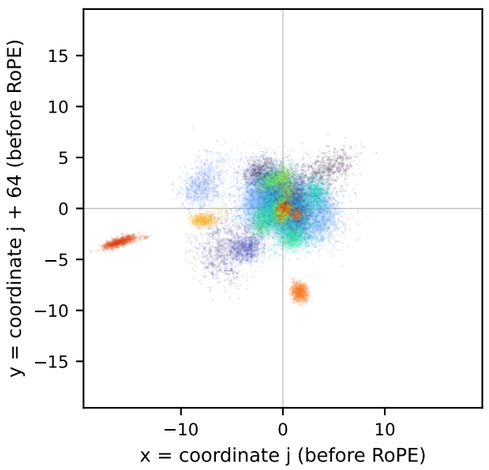
L11 H2



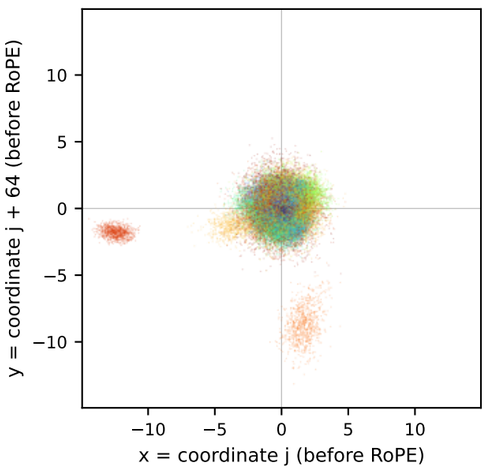
L11 H3



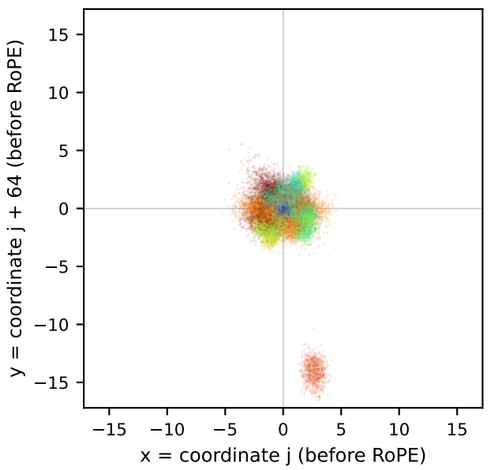
L11 H4



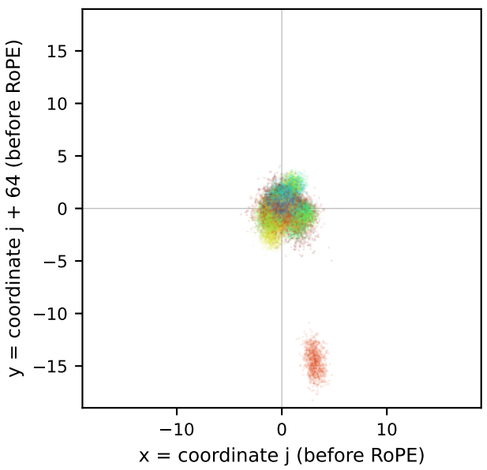
L11 H5



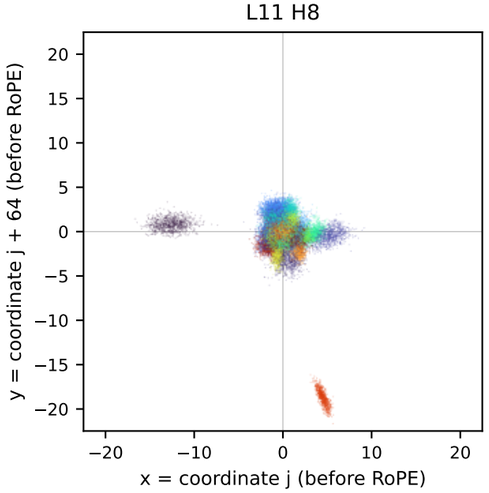
L11 H6



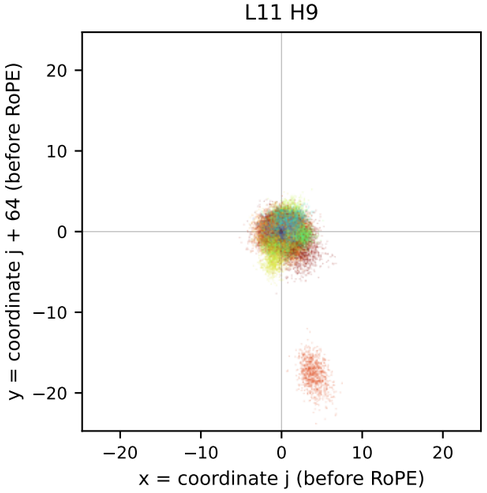
L11 H7



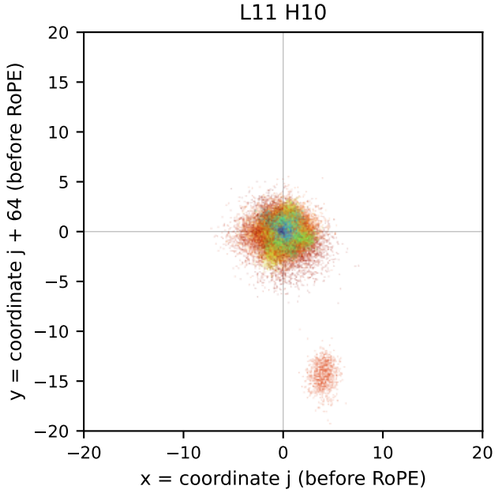
L11 H8



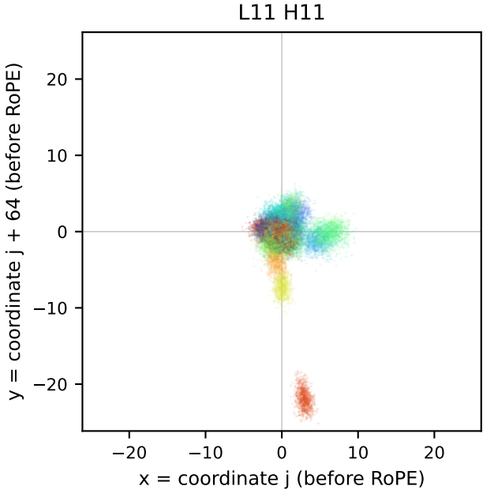
L11 H9

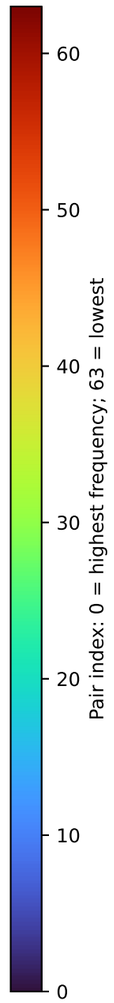


L11 H10



L11 H11





Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 12 · axes fitted separately for each head

L12 H0

L12 H1

L12 H2

L12 H3

L12 H4

L12 H5

L12 H6

L12 H7

L12 H8

L12 H9

L12 H10

L12 H11

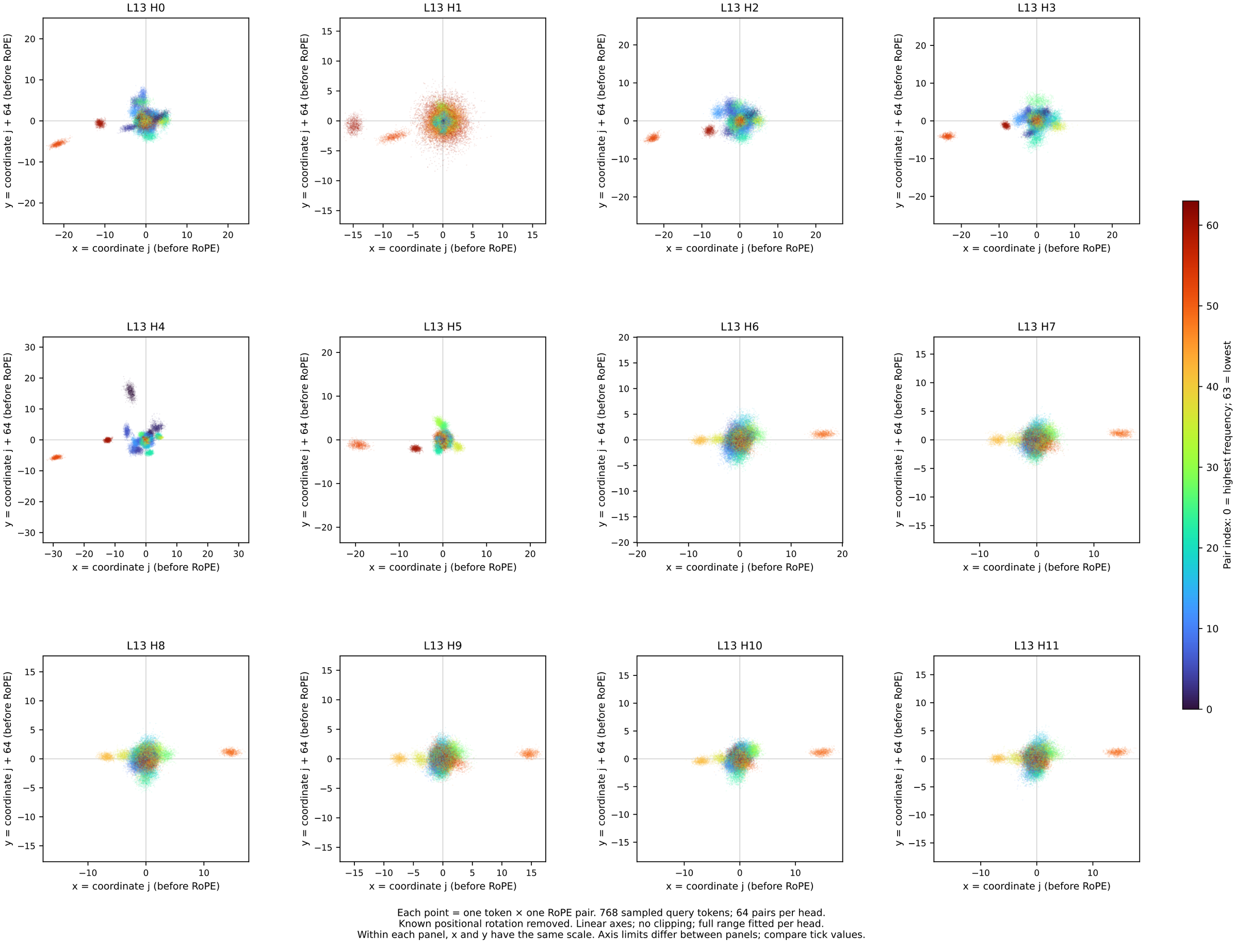
y = coordinate j + 64 (before RoPE)

x = coordinate j (before RoPE)

Pair index: 0 = highest frequency; 63 = lowest

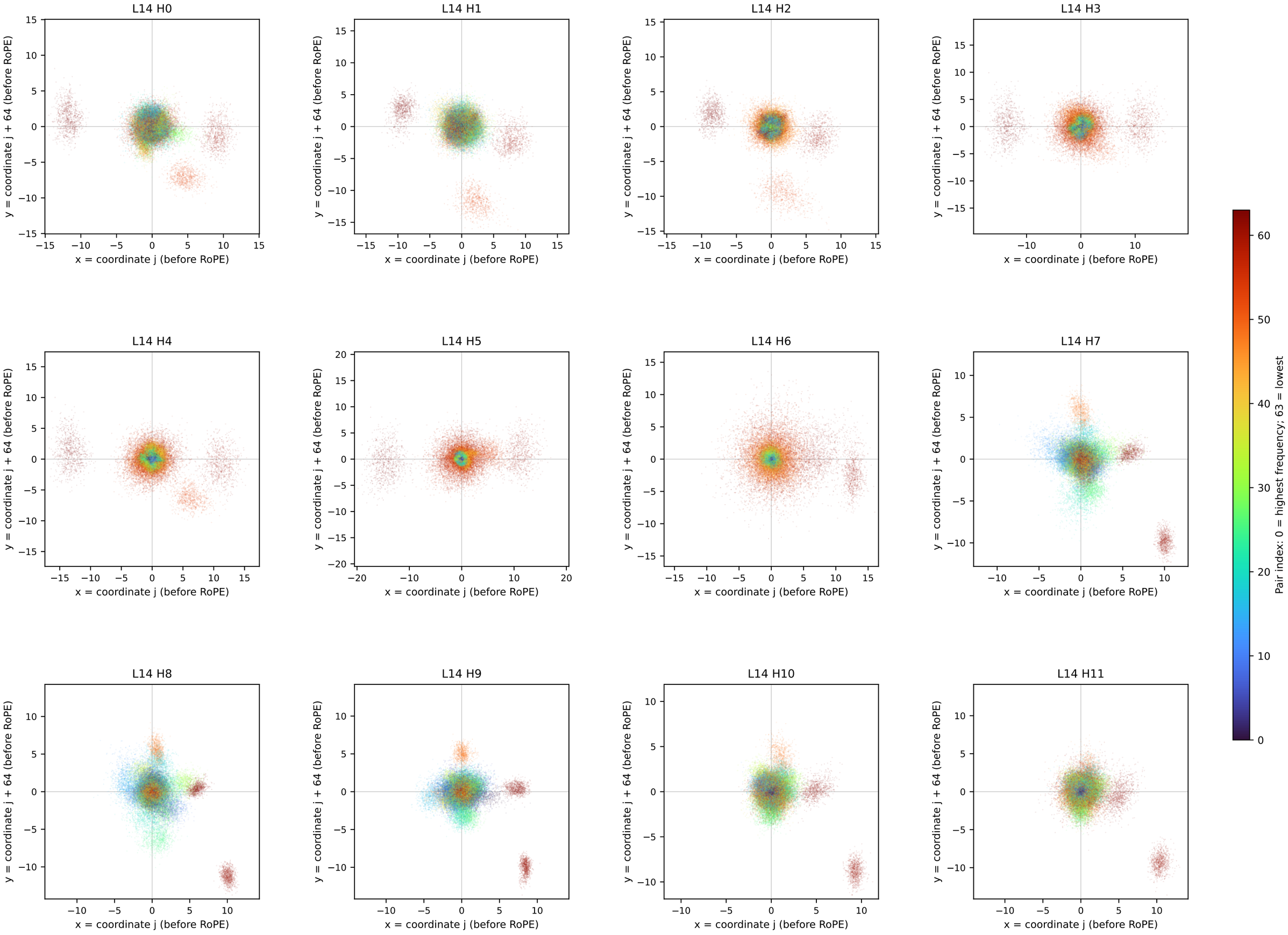
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 13 · axes fitted separately for each head



Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

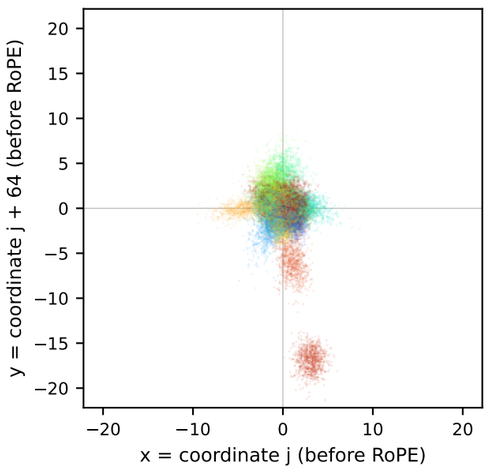
Qwen queries · layer 14 · axes fitted separately for each head



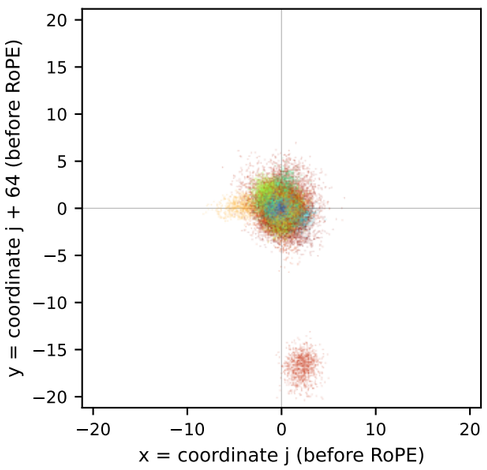
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 15 · axes fitted separately for each head

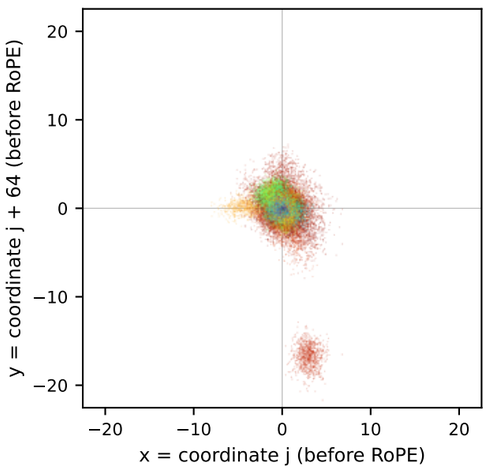
L15 H0



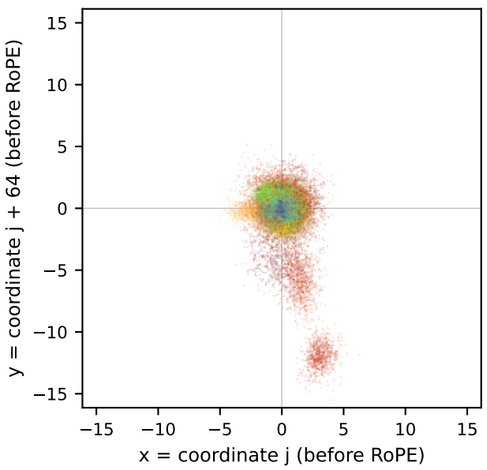
L15 H1



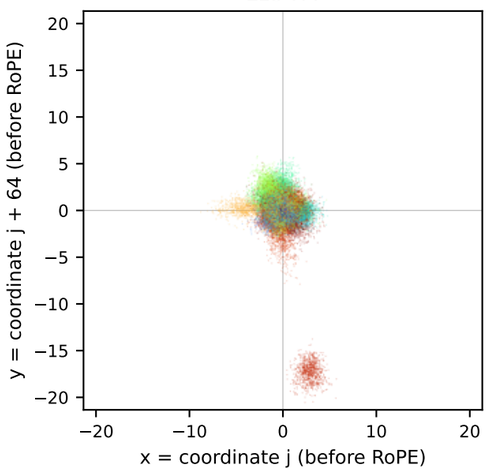
L15 H2



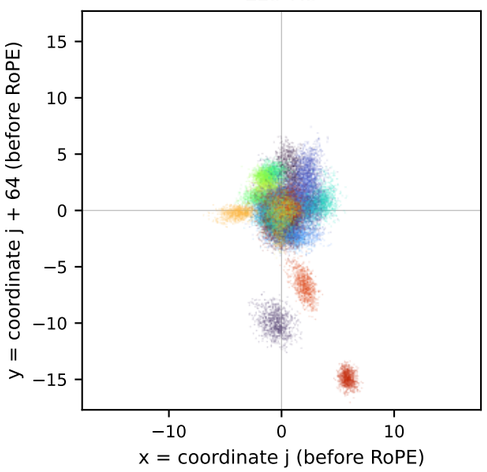
L15 H3



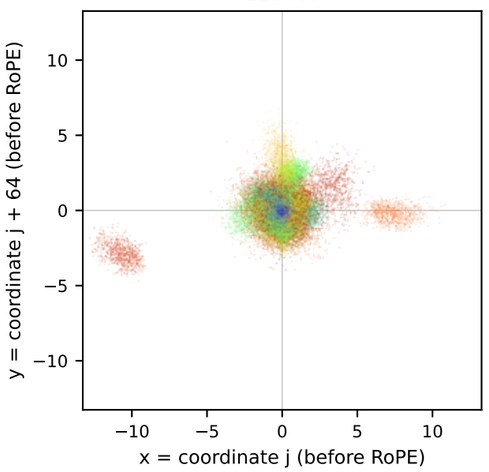
L15 H4



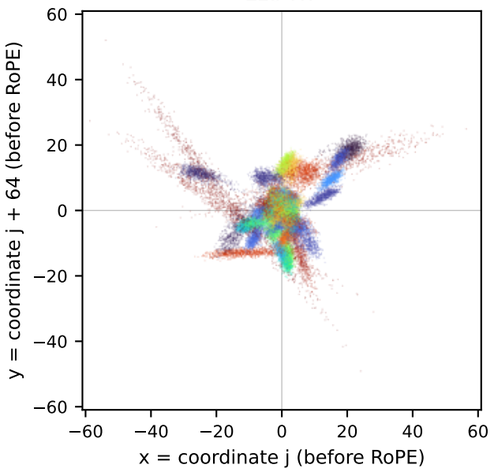
L15 H5



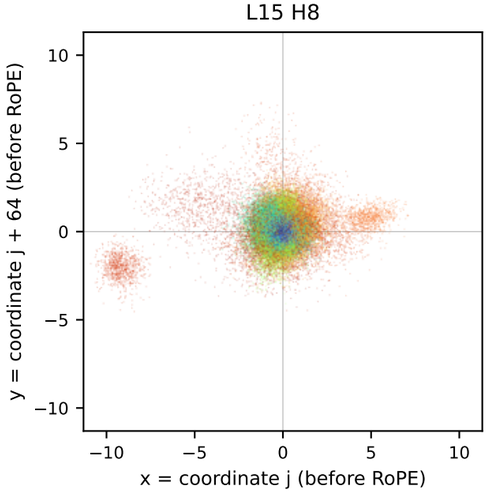
L15 H6



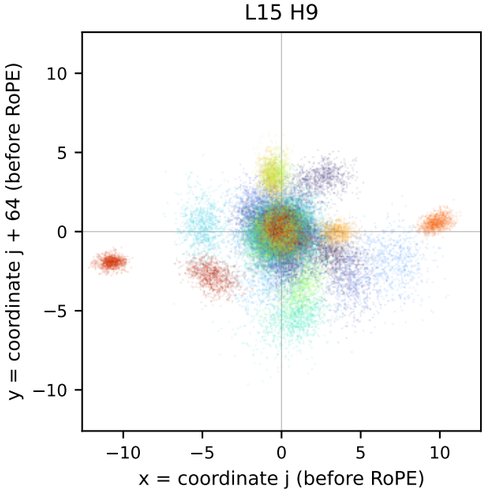
L15 H7



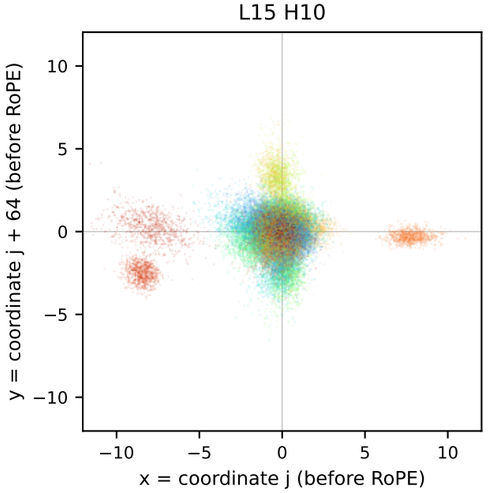
L15 H8



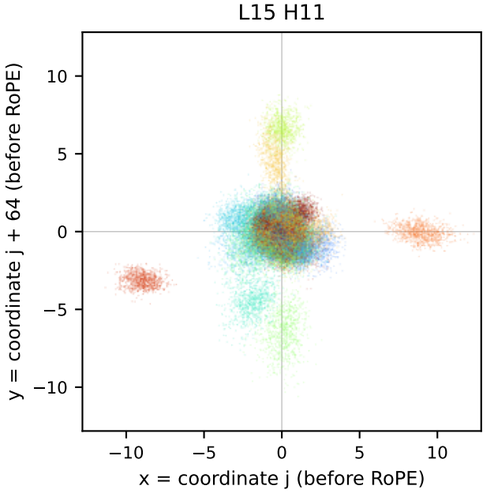
L15 H9

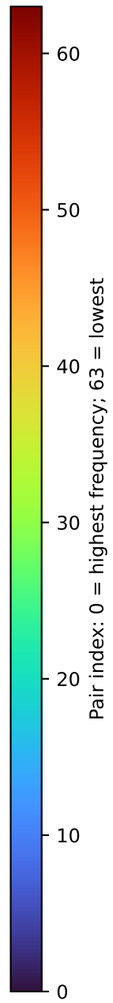


L15 H10



L15 H11

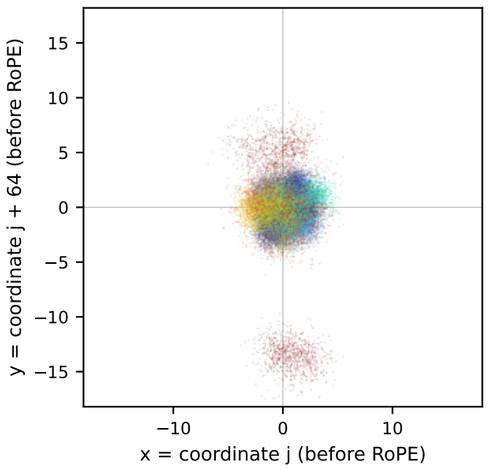




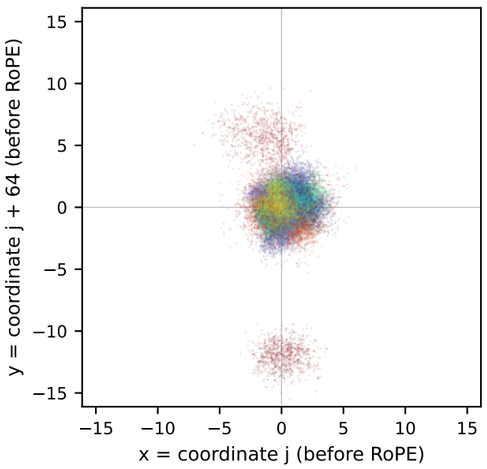
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 16 · axes fitted separately for each head

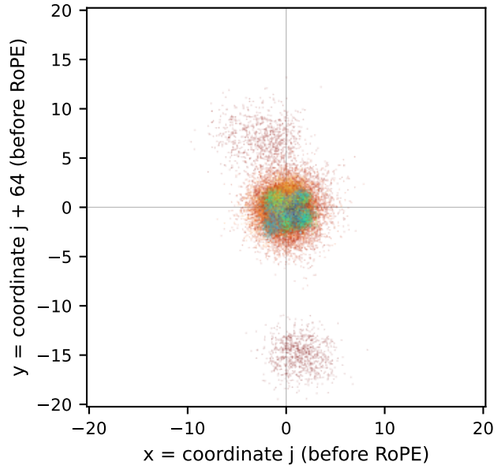
L16 H0



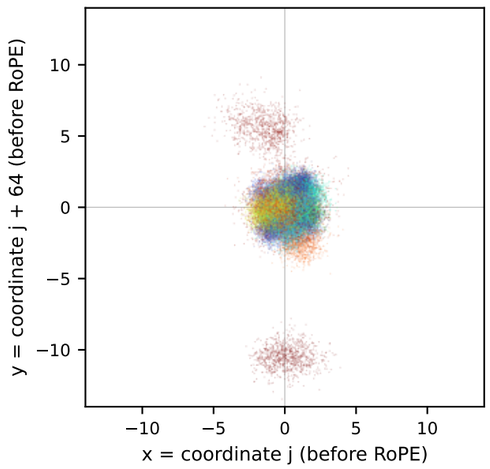
L16 H1



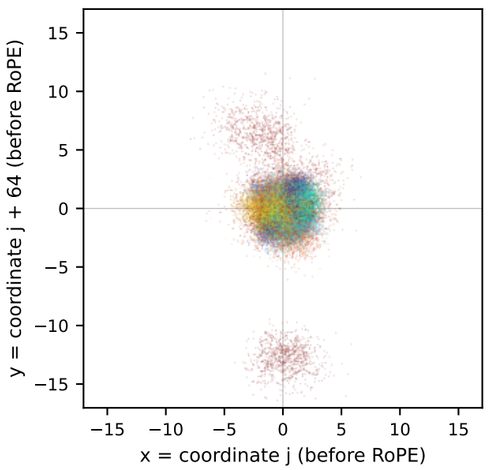
L16 H2



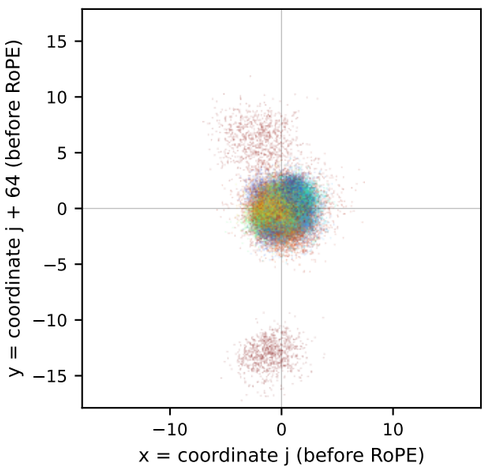
L16 H3



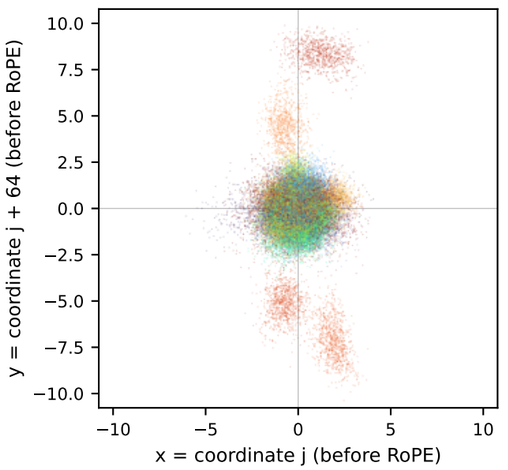
L16 H4



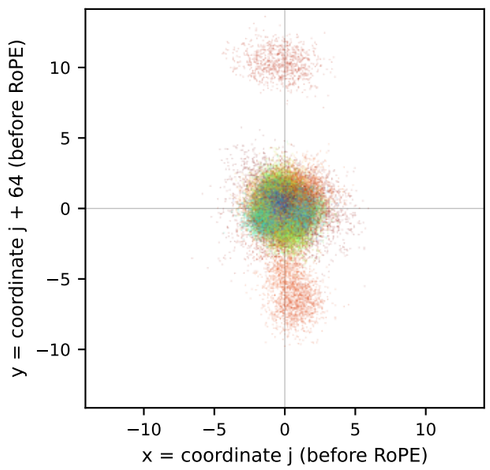
L16 H5



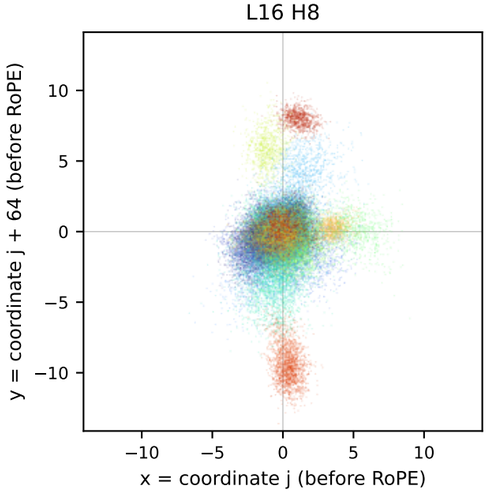
L16 H6



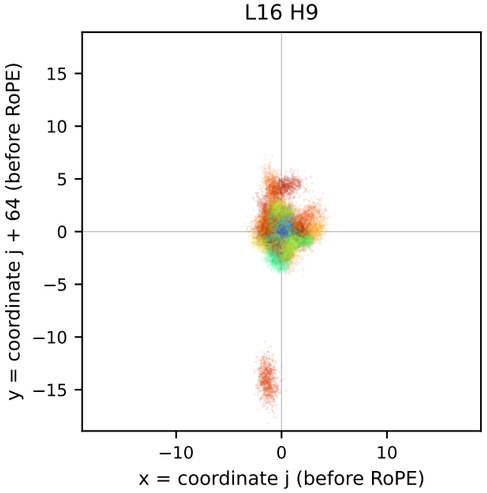
L16 H7



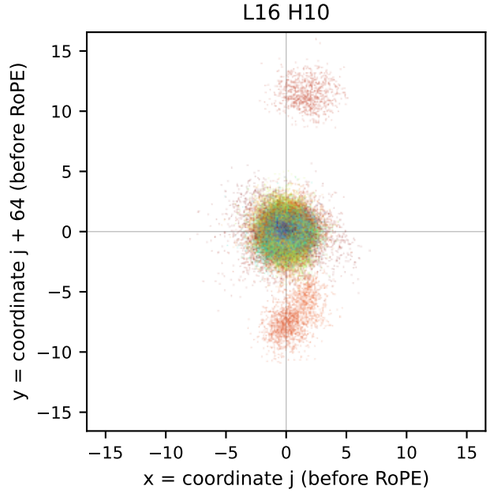
L16 H8



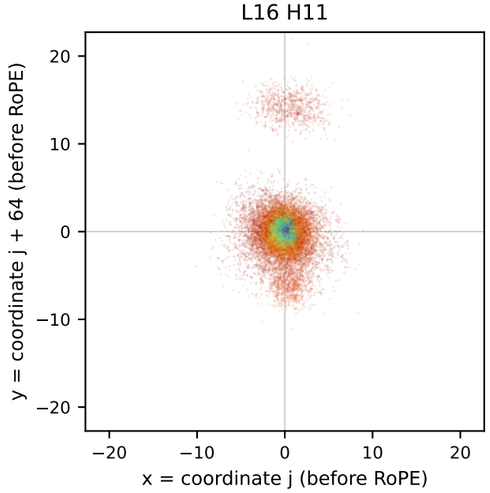
L16 H9

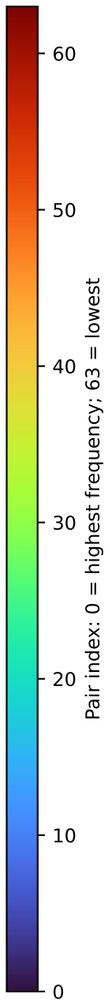


L16 H10



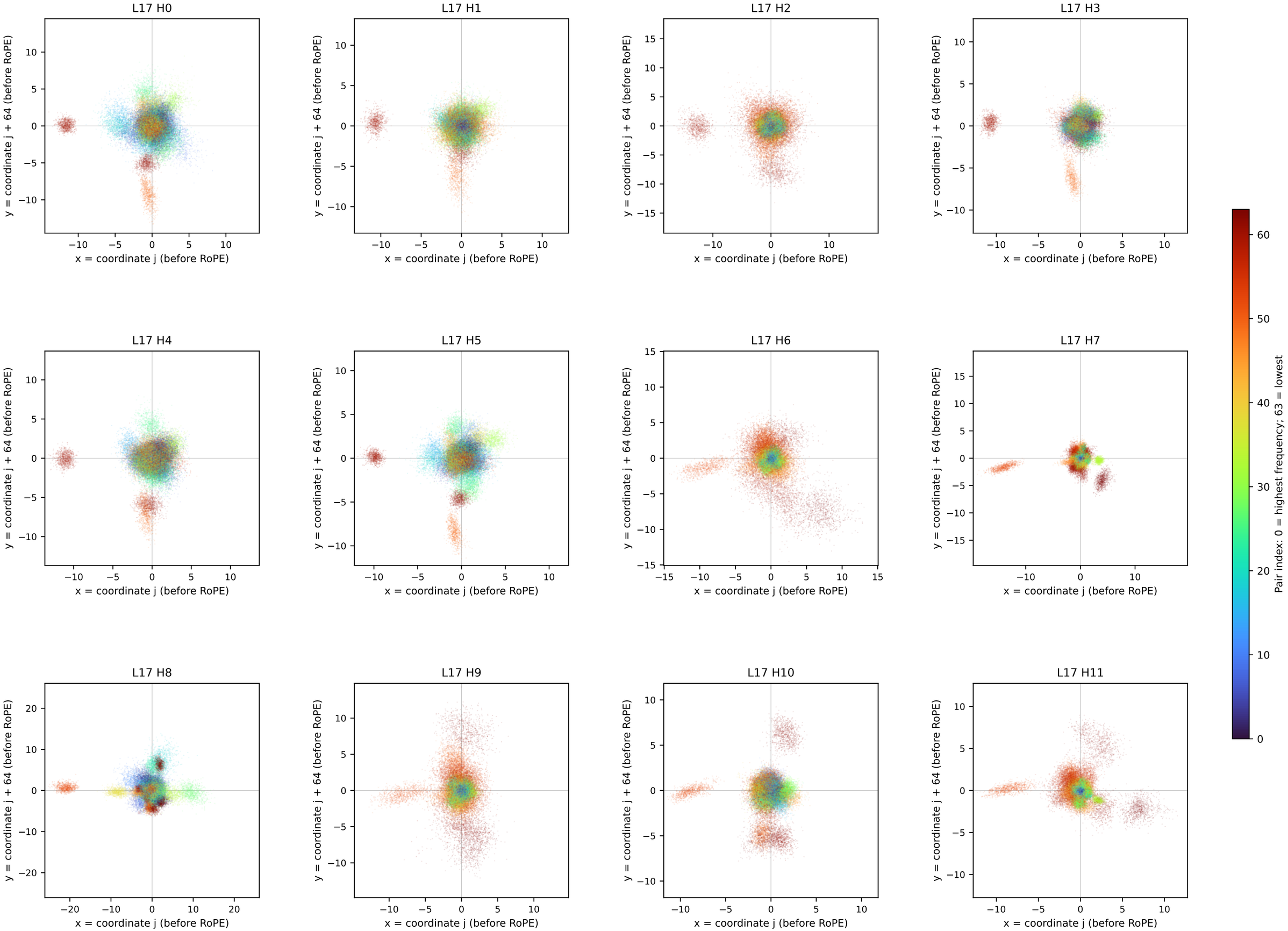
L16 H11





Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

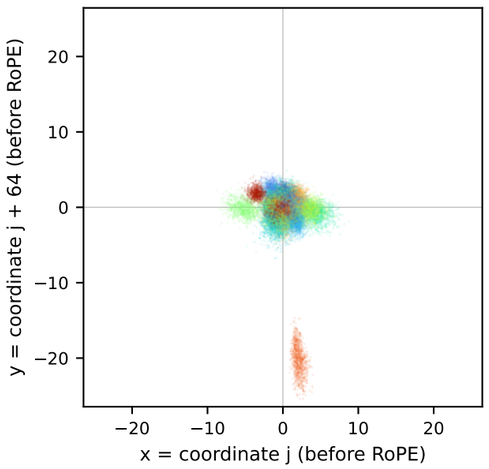
Qwen queries · layer 17 · axes fitted separately for each head



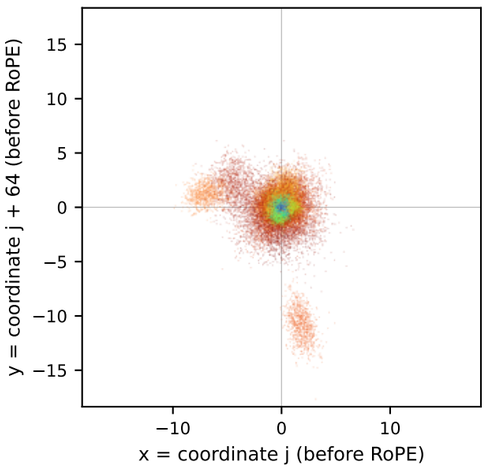
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 18 · axes fitted separately for each head

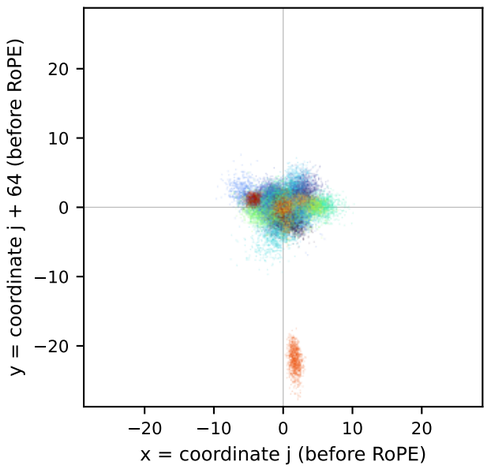
L18 H0



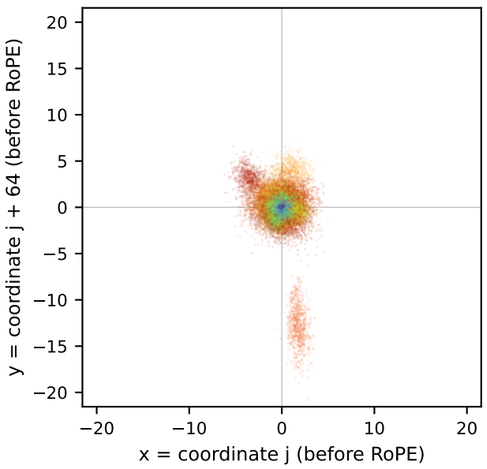
L18 H1



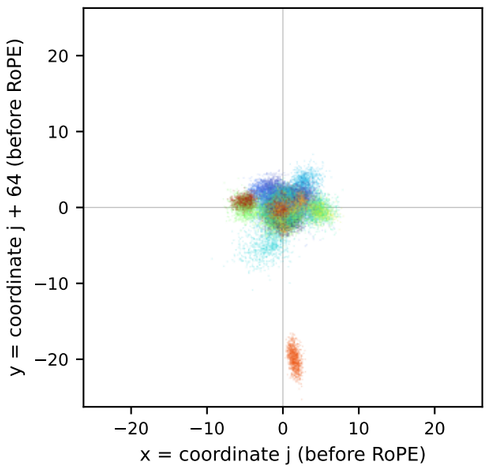
L18 H2



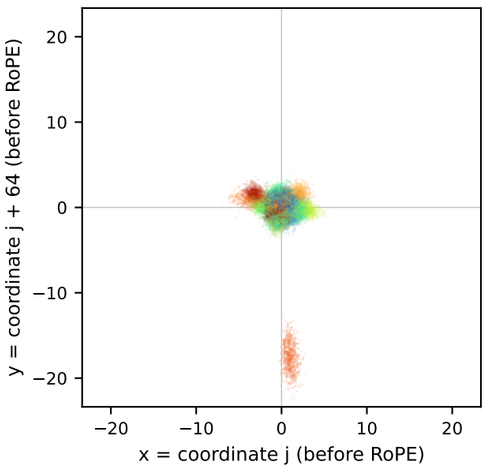
L18 H3



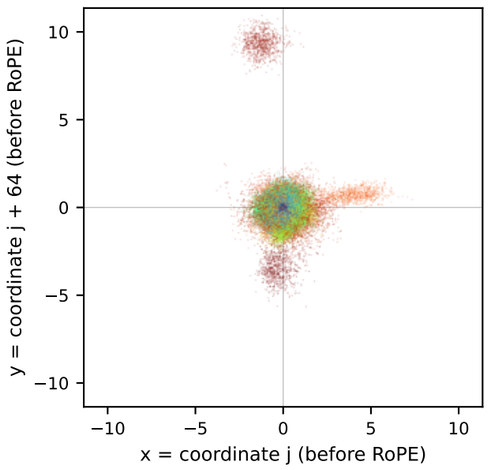
L18 H4



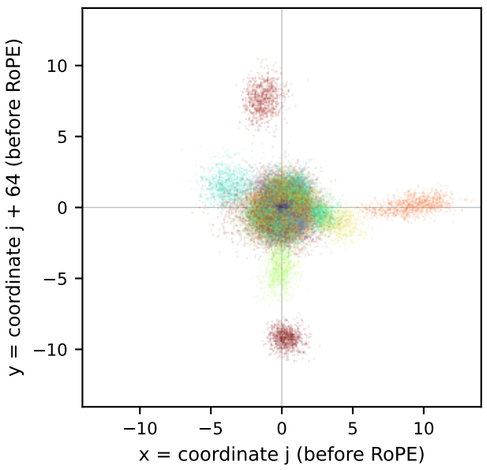
L18 H5



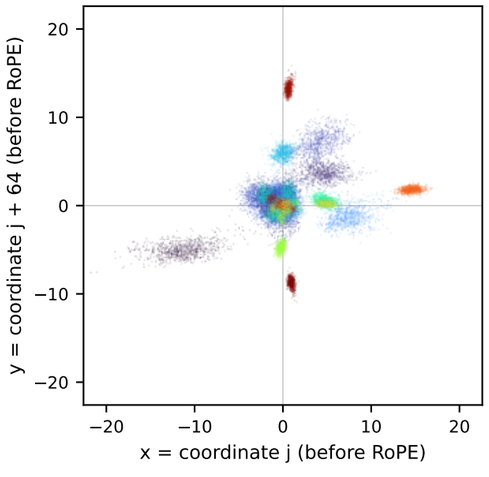
L18 H6



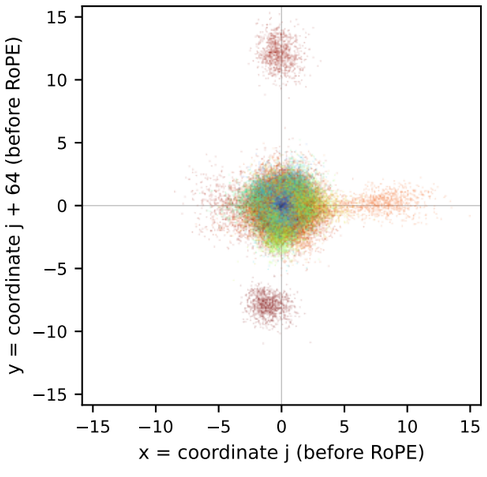
L18 H7



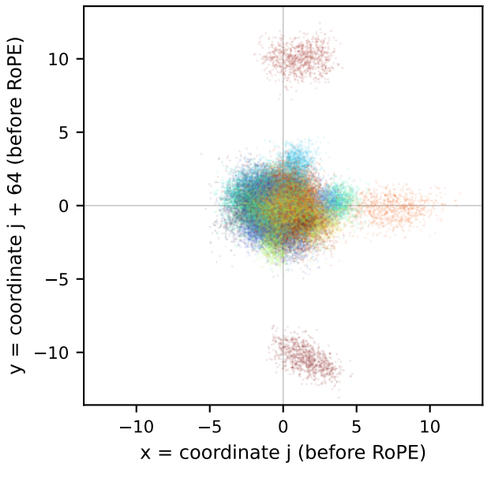
L18 H8



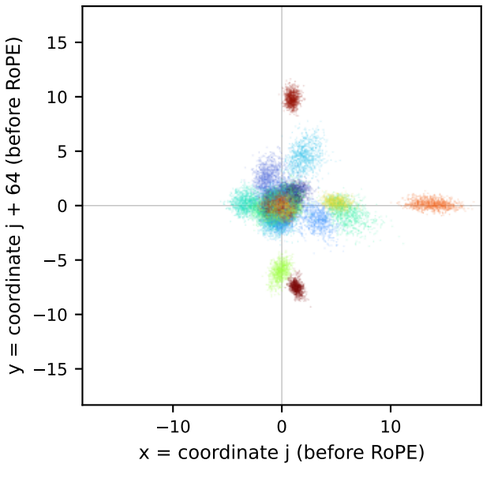
L18 H9

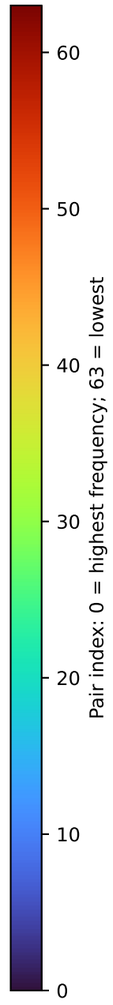


L18 H10



L18 H11

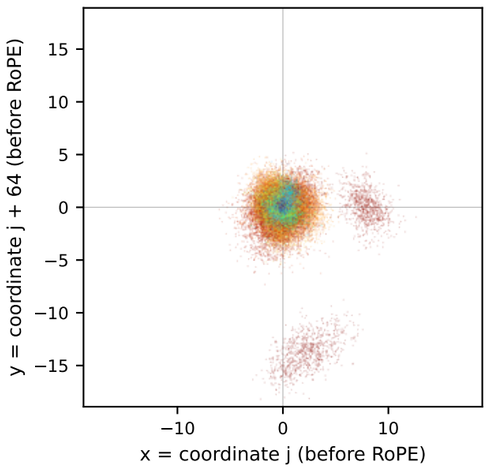




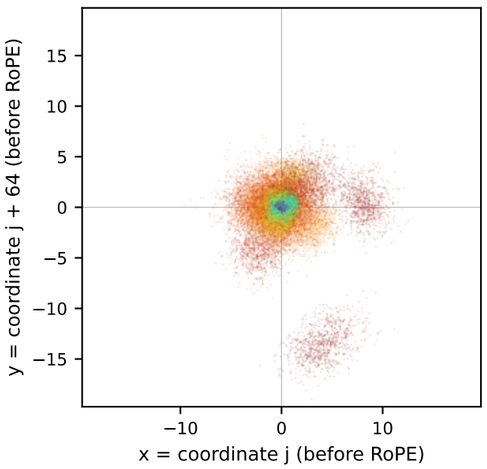
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Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 19 · axes fitted separately for each head

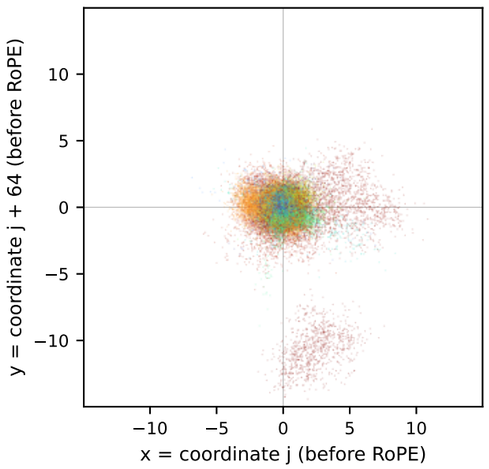
L19 H0



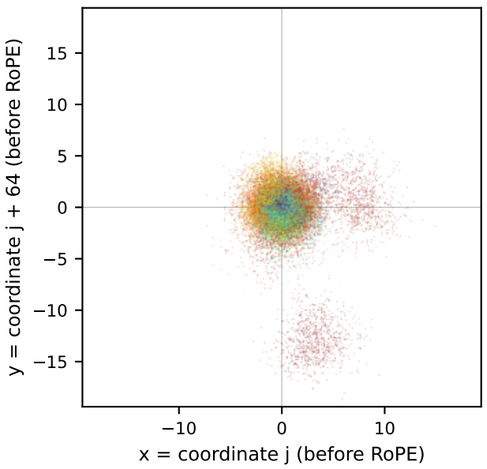
L19 H1



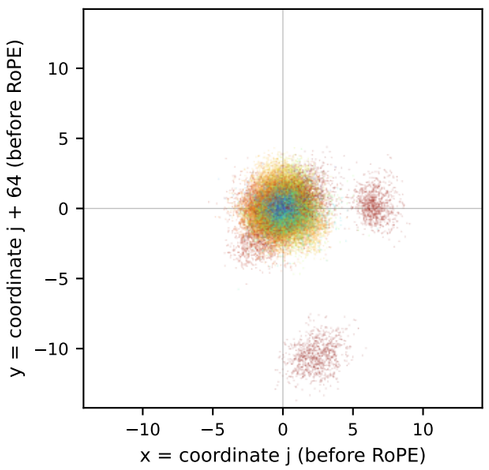
L19 H2



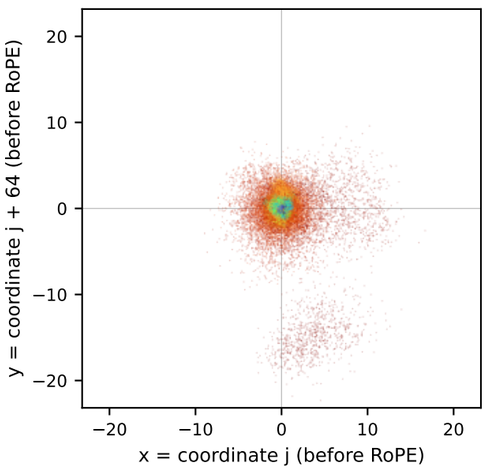
L19 H3



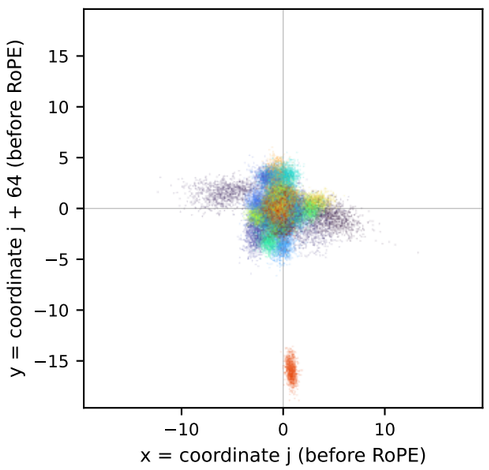
L19 H4



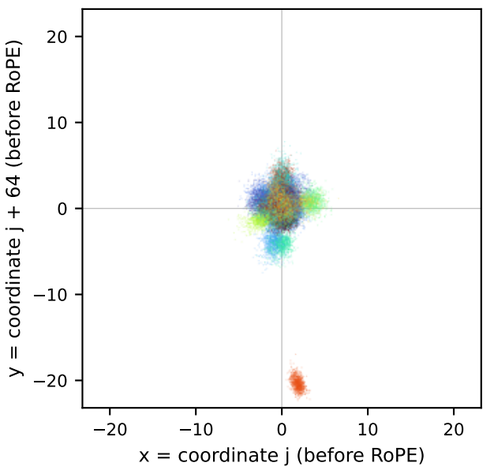
L19 H5



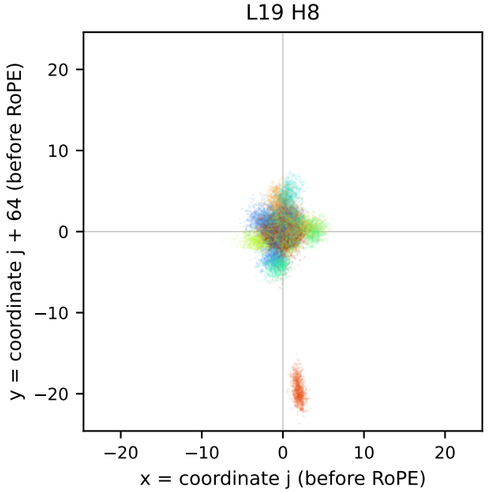
L19 H6



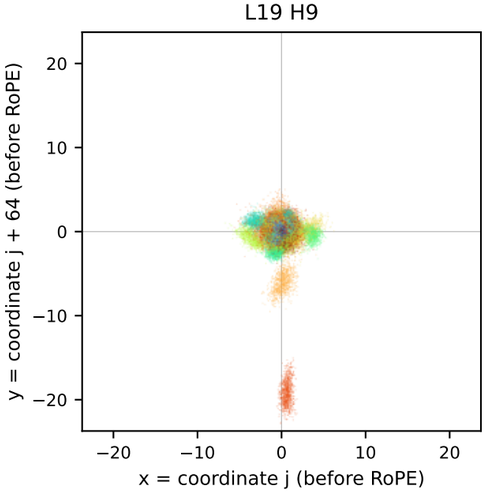
L19 H7



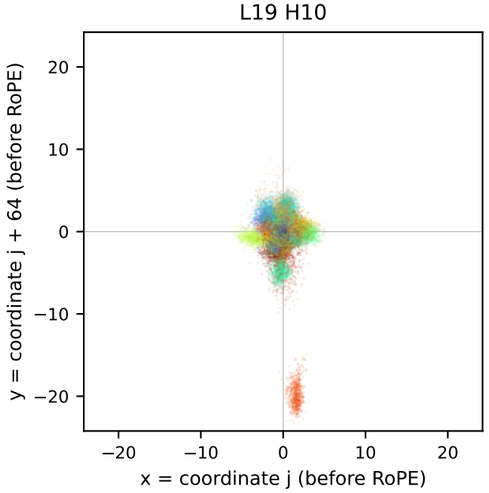
L19 H8



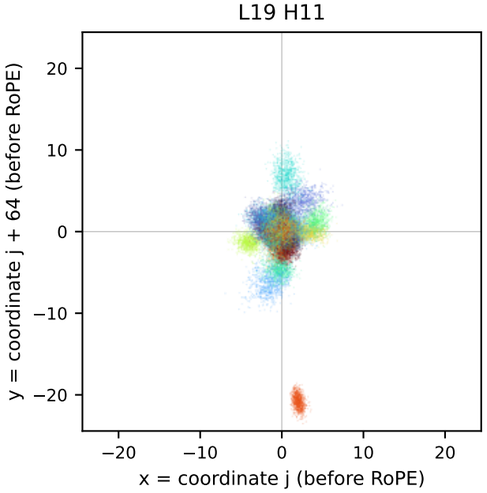
L19 H9

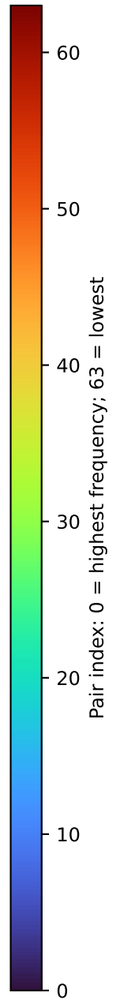


L19 H10



L19 H11

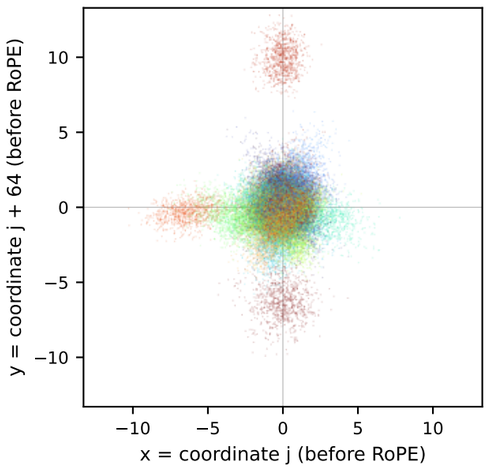




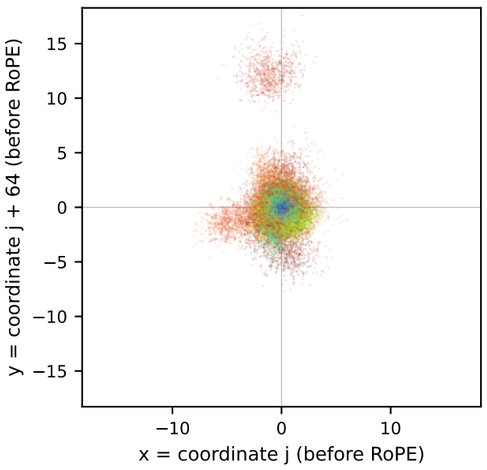
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Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 20 · axes fitted separately for each head

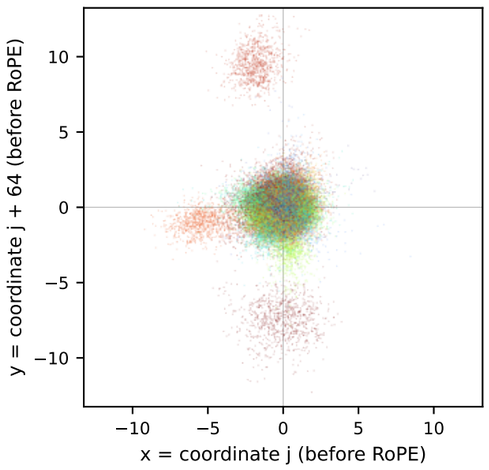
L20 H0



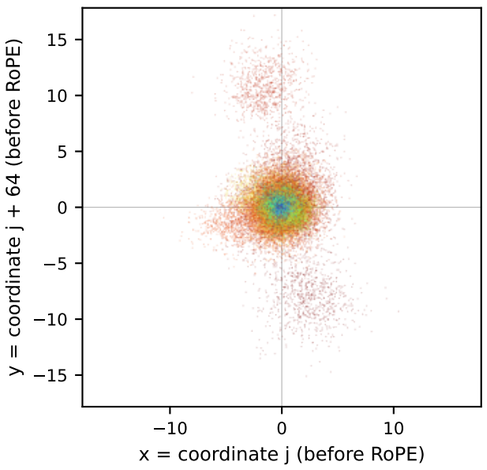
L20 H1



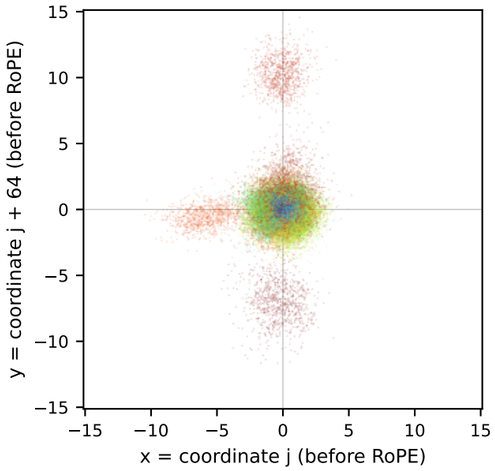
L20 H2



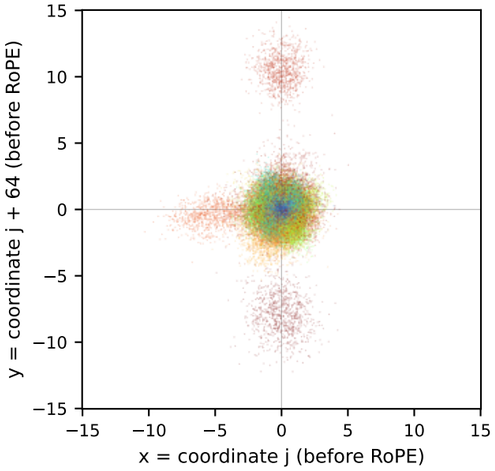
L20 H3



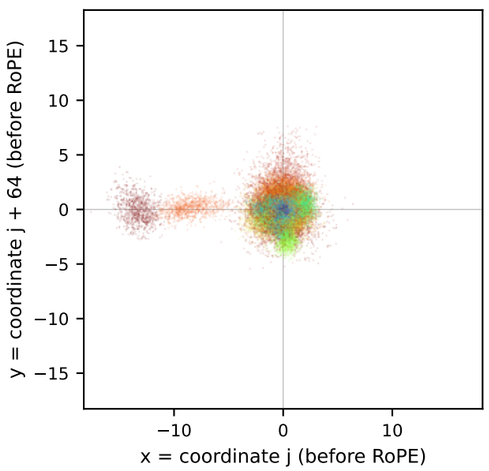
L20 H4



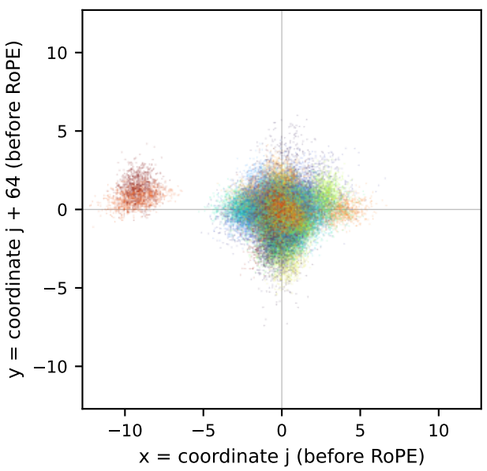
L20 H5



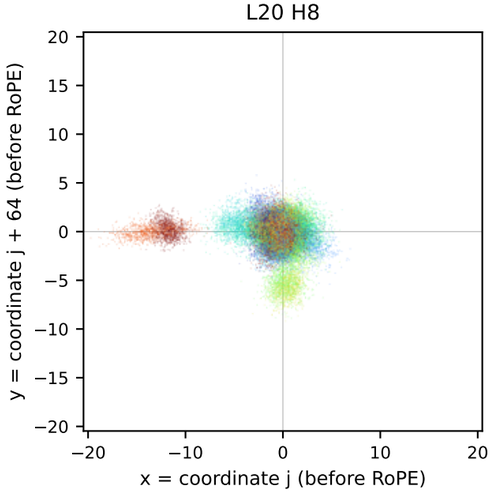
L20 H6



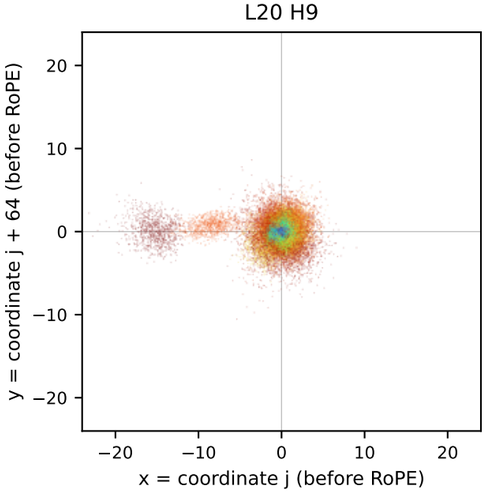
L20 H7



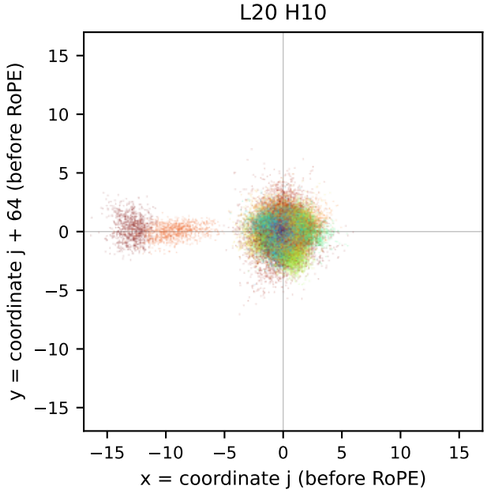
L20 H8



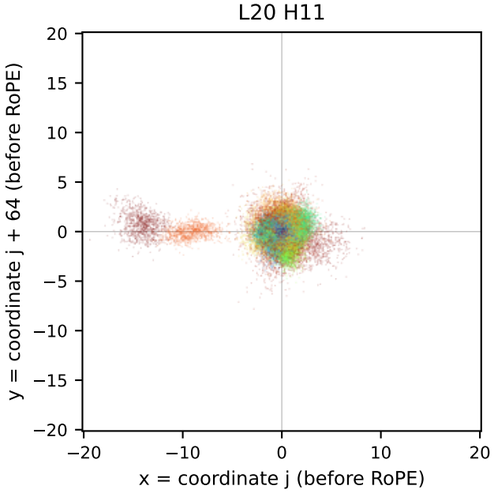
L20 H9

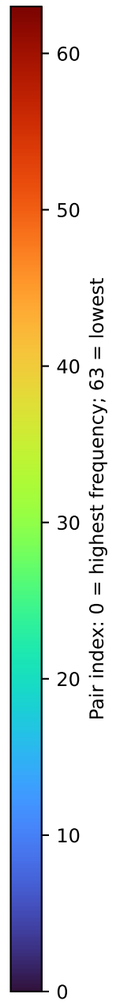


L20 H10



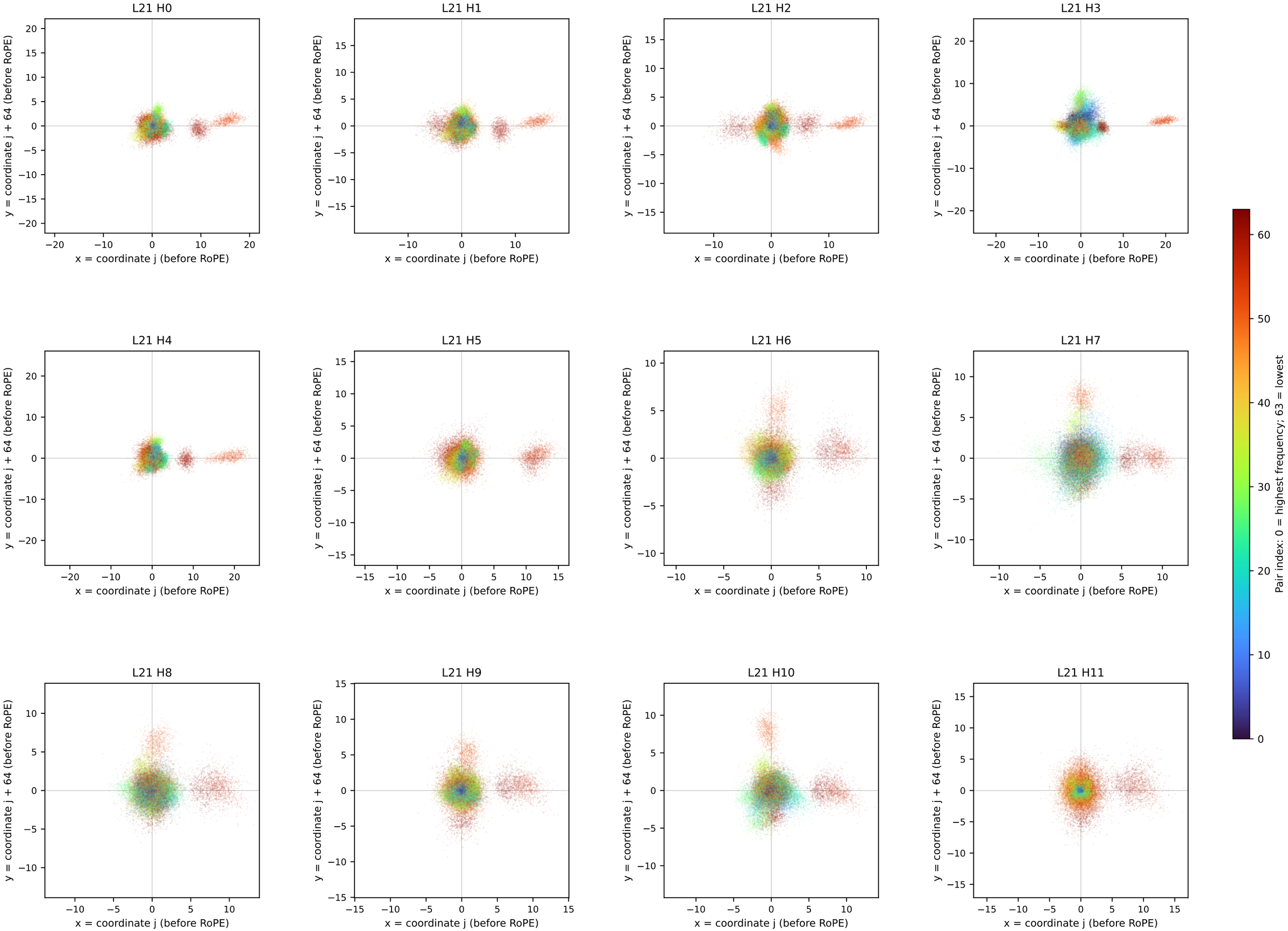
L20 H11





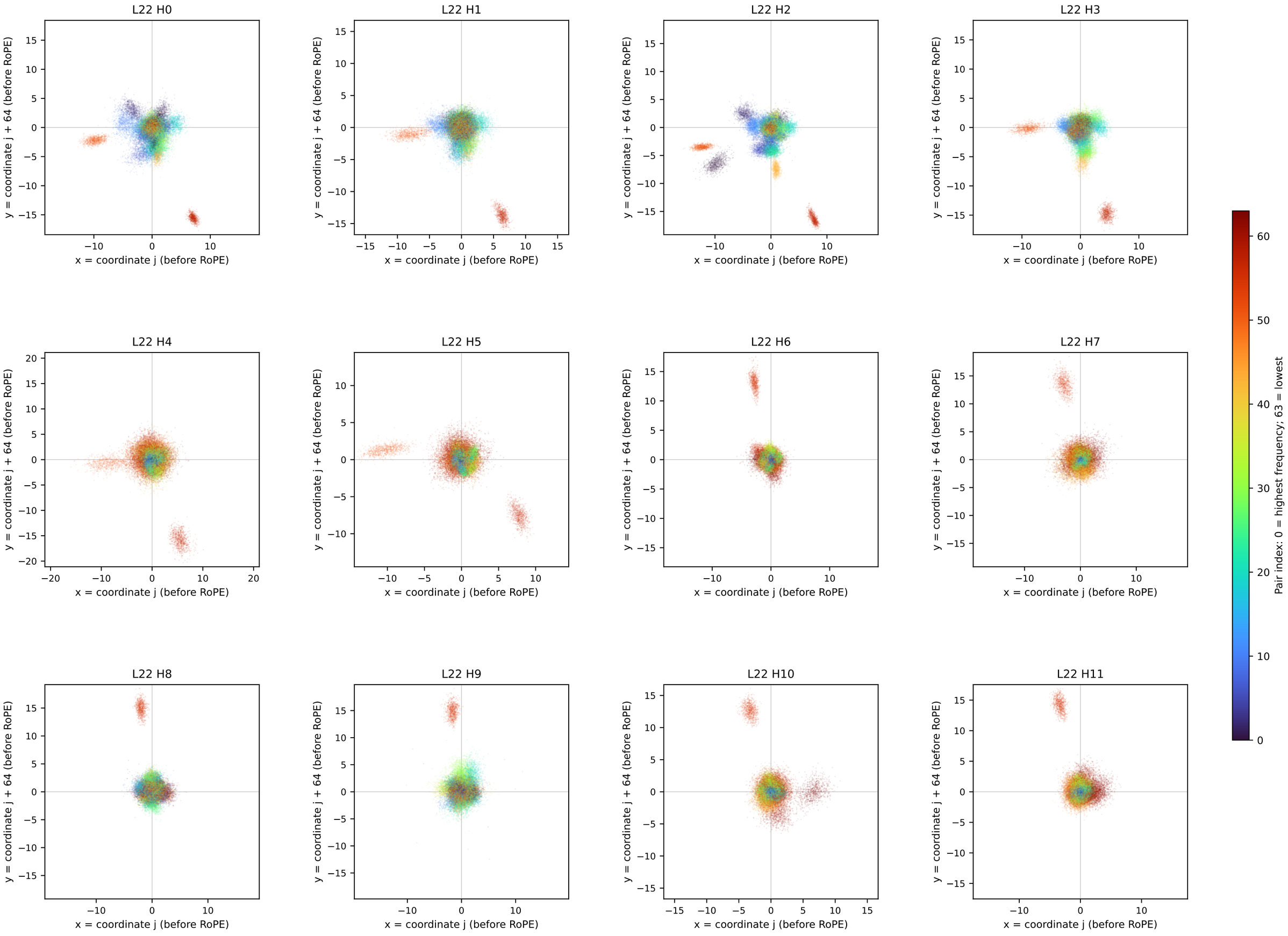
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 21 · axes fitted separately for each head



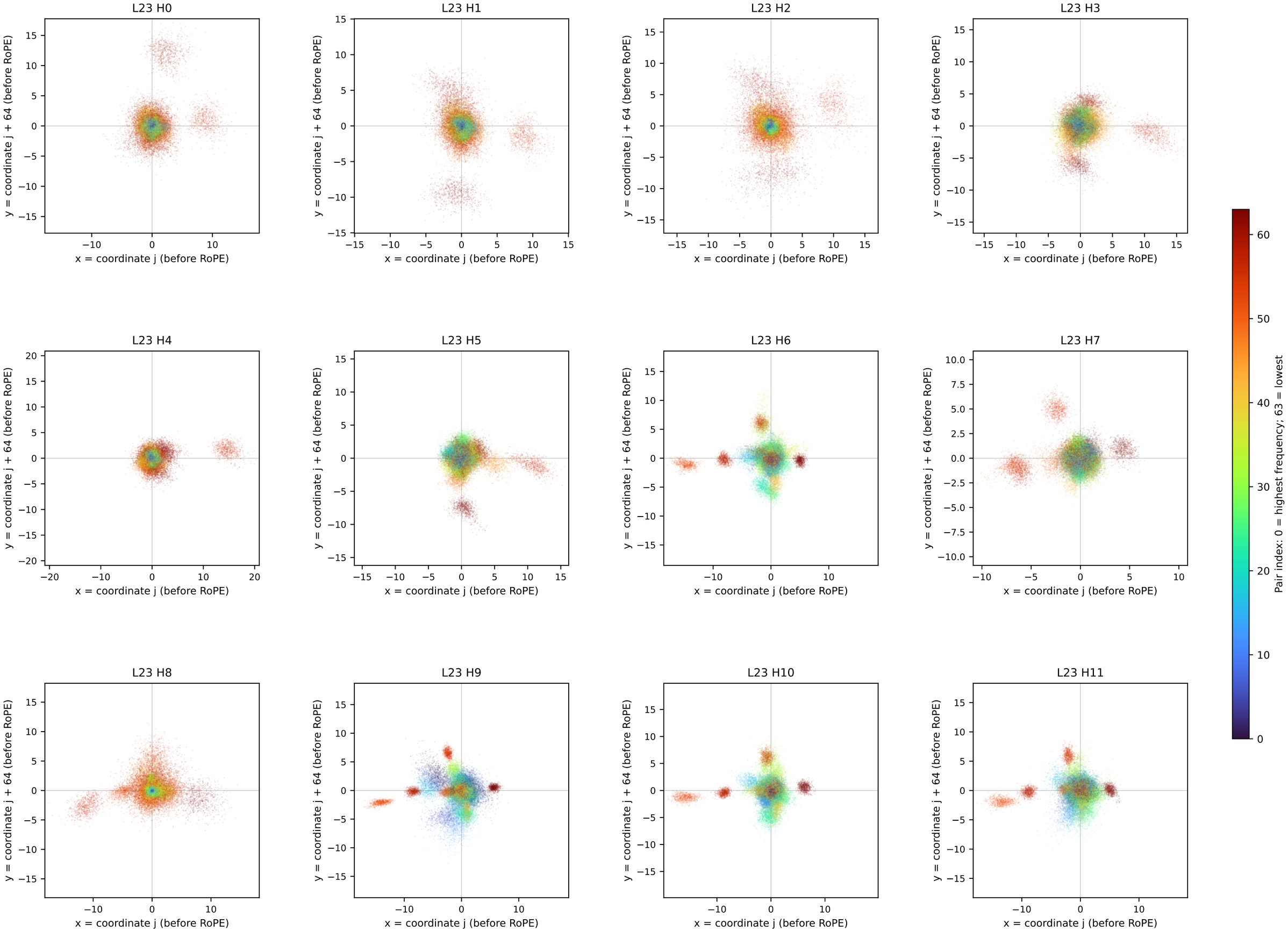
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 22 · axes fitted separately for each head



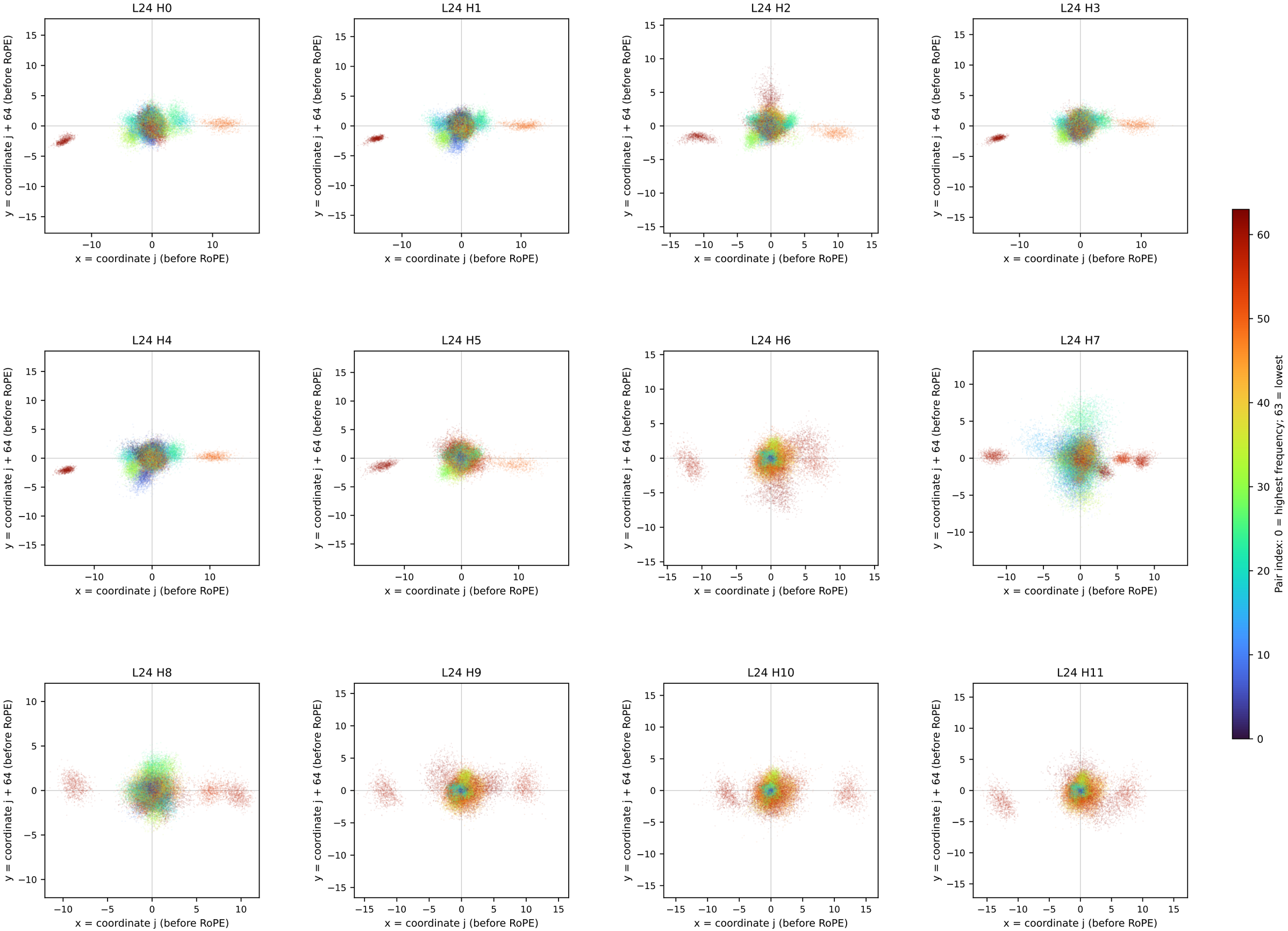
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 23 · axes fitted separately for each head



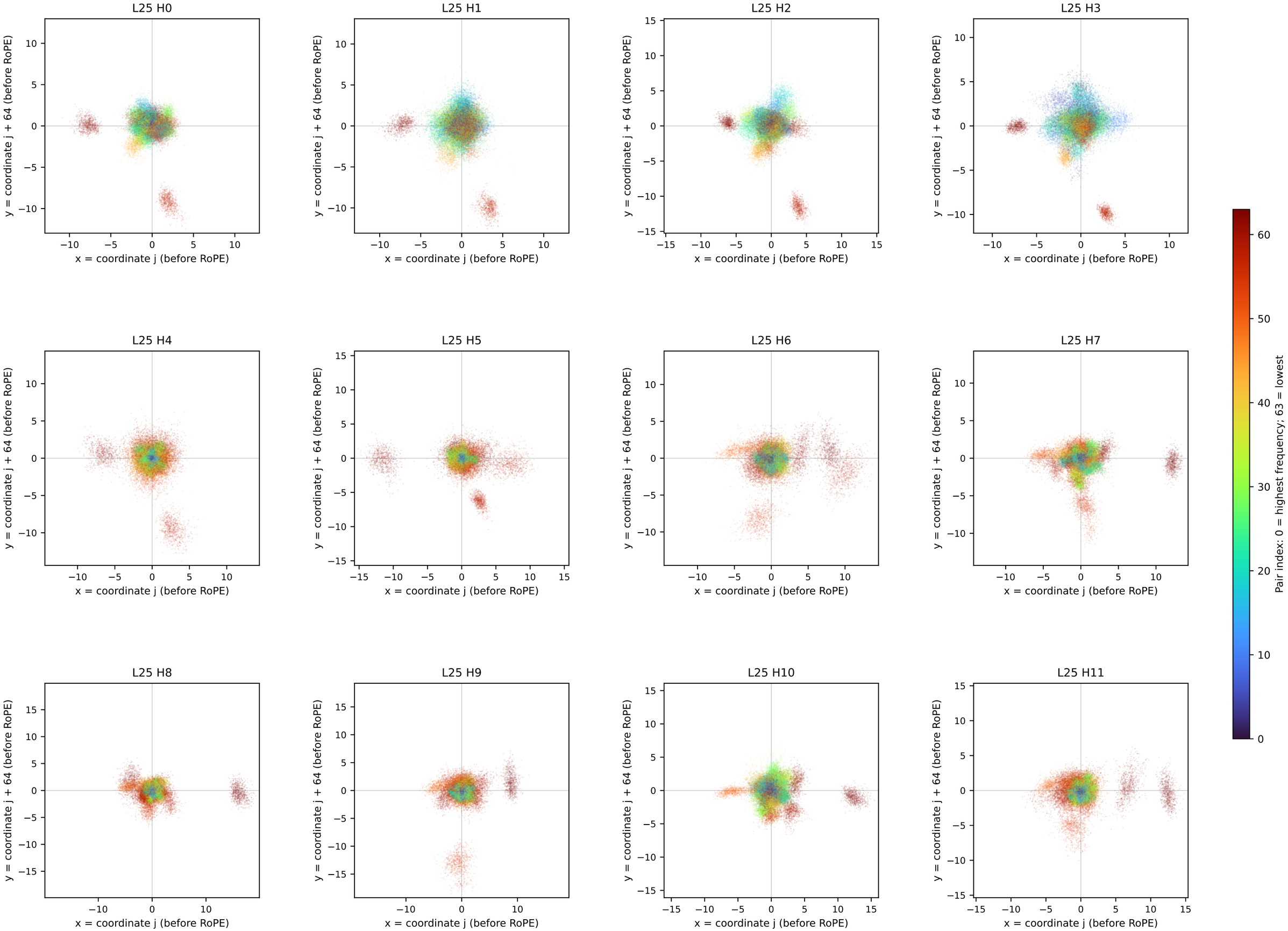
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 24 · axes fitted separately for each head



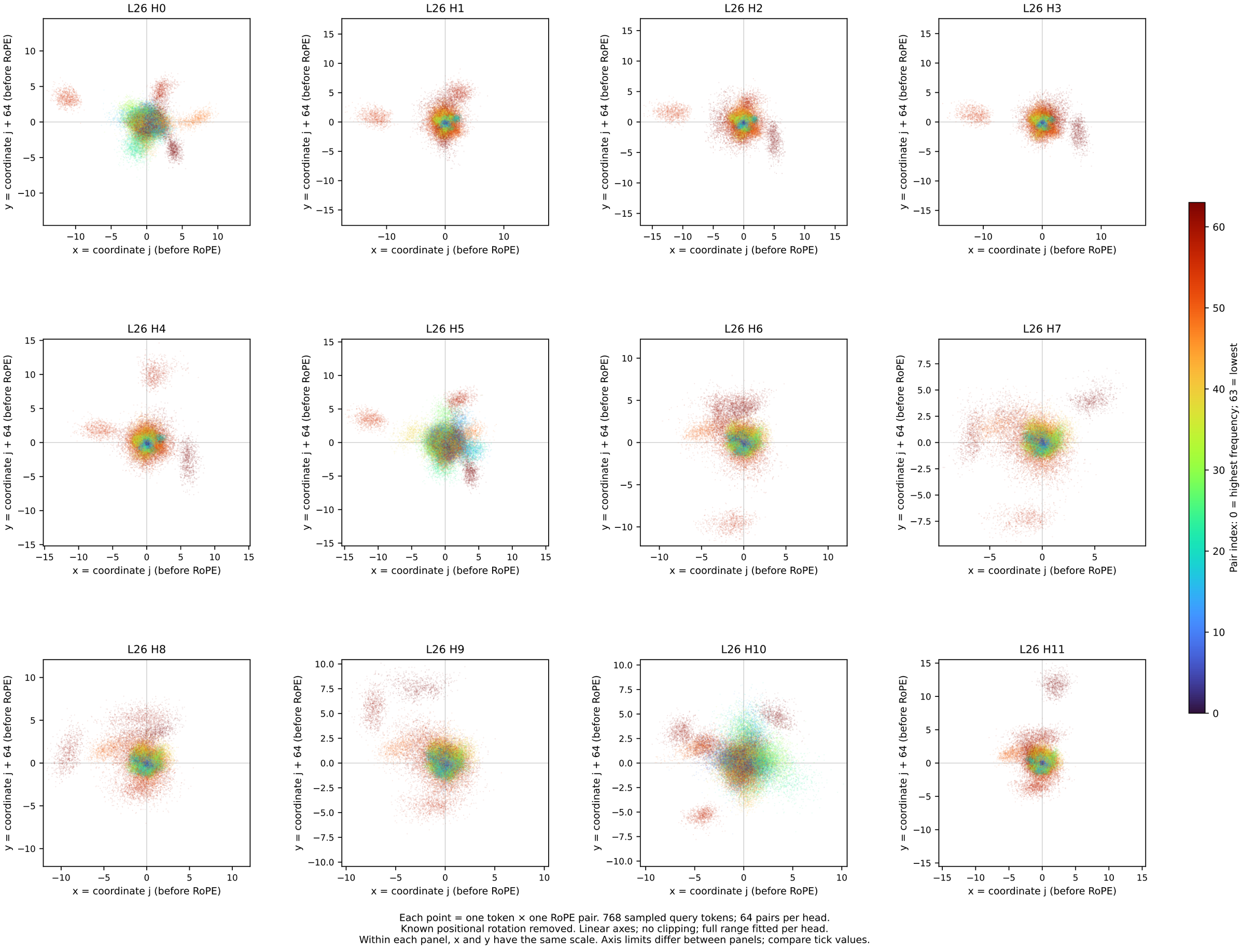
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 25 · axes fitted separately for each head



Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

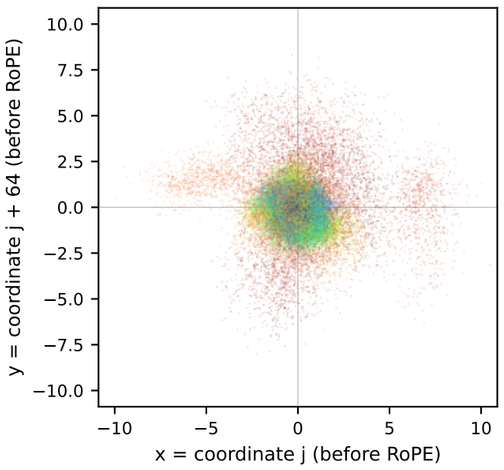
Qwen queries · layer 26 · axes fitted separately for each head



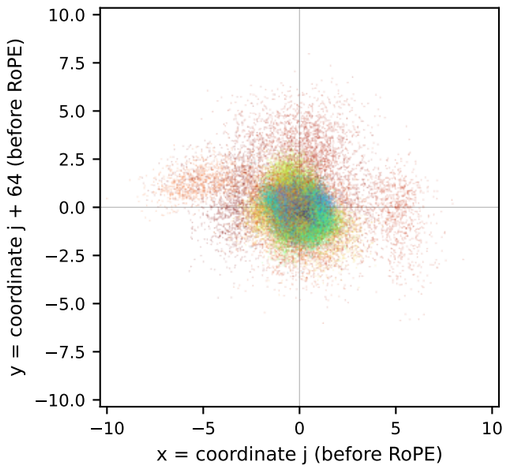
Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.

Qwen queries · layer 27 · axes fitted separately for each head

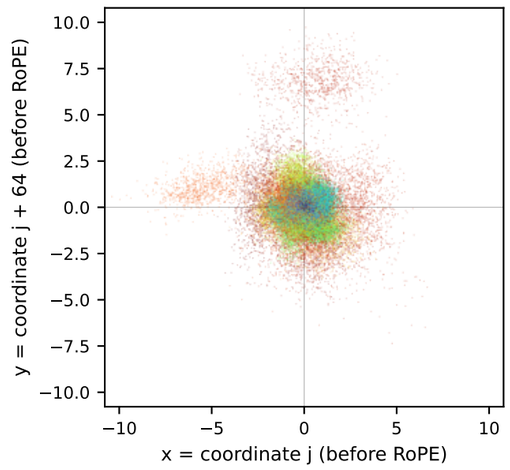
L27 H0



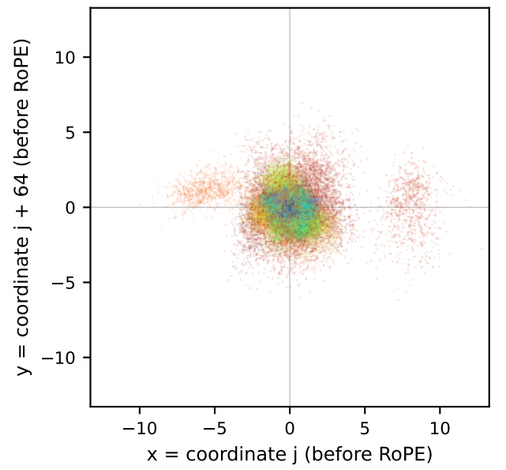
L27 H1



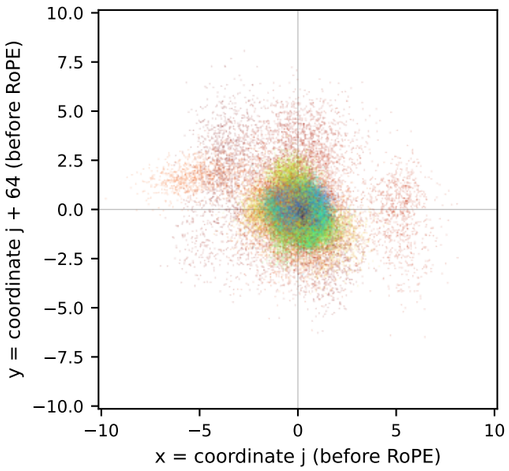
L27 H2



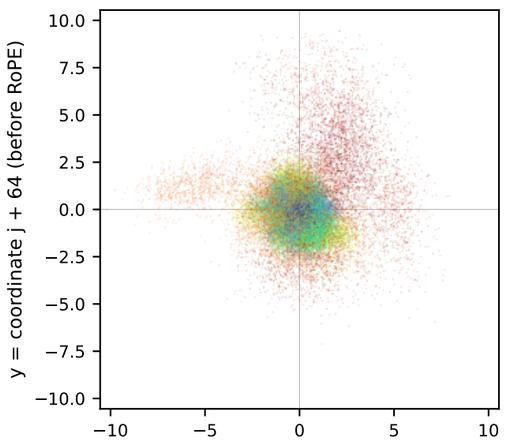
L27 H3



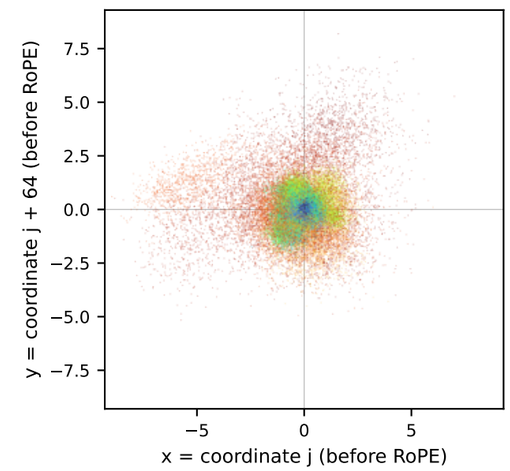
L27 H4



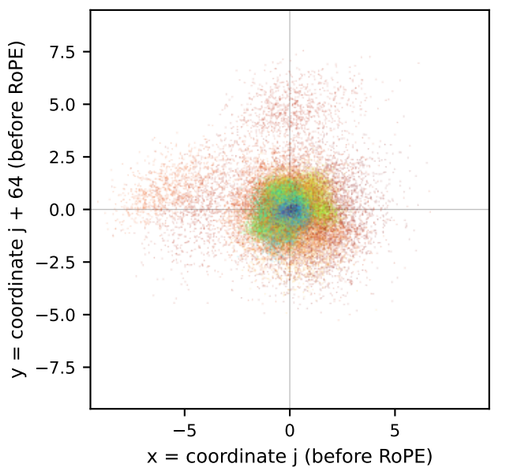
L27 H5



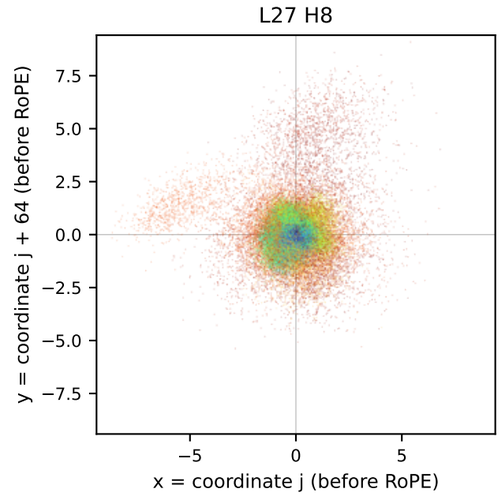
L27 H6



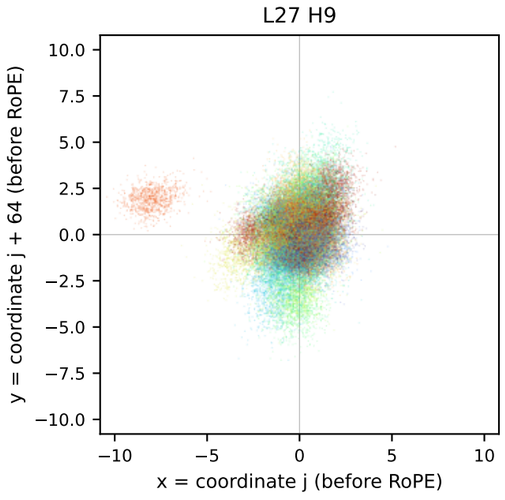
L27 H7



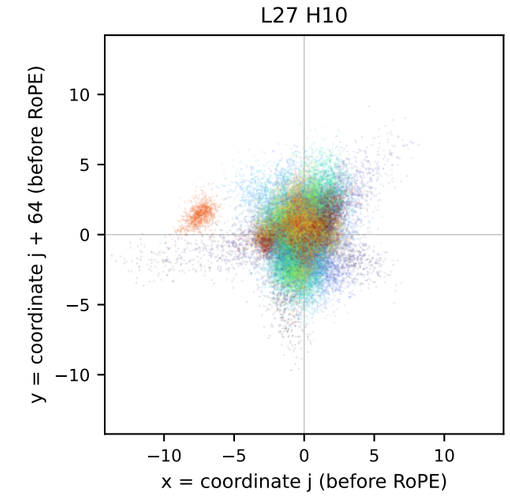
L27 H8



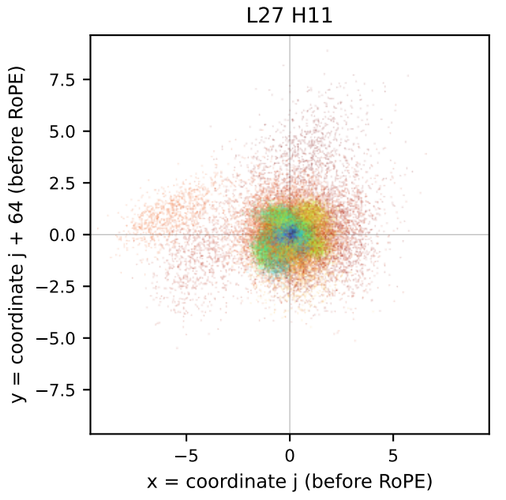
L27 H9

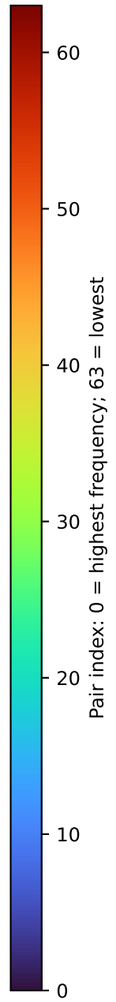


L27 H10



L27 H11





Each point = one token × one RoPE pair. 768 sampled query tokens; 64 pairs per head.
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